Journal of The
Electrochemical Society

OPEN ACCESS

How do Depth of Discharge, C-rate and Calendar Age Affect Capacity Retention, Impedance Growth, the Electrodes, and the Electrolyte in Li-Ion Cells?

To cite this article: Roby Gauthier *et al* 2022 *J. Electrochem. Soc.* **169** 020518

View the article online for updates and enhancements.

You may also like

- Calendar Aging and Gas Generation in Commercial Graphite/NMC-LMO Lithium-Ion Pouch Cell
Z. Mao, M. Farkhondeh, M. Pritzker et al.
- Electrochemical Cycle-Life Characterization of High Energy Lithium-Ion Cells with Thick $Li(Ni_{0.6}Mn_{0.2}Co_{0.2})O_2$ and Graphite Electrodes
Yongjun Leng, Shanhai Ge, Dan Marple et al.
- (Invited) Effect of Aging on Mechanical Properties of Lithium Ion Cell Components
Zenan Wu, Lei Cao, Julia Hartig et al.

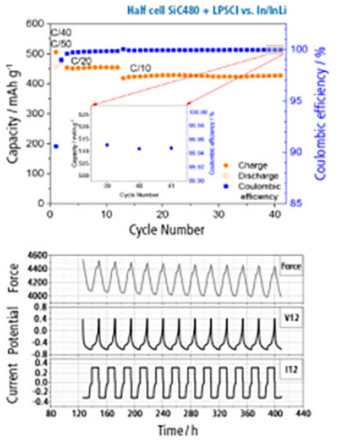
The New PAT-Cell-Solid!

EL-CELL® electrochemical test equipment

Cycle Solid-State Batteries Under Controlled Pressure of up to 300 MPa (6 mm Diameter)!



- Adjust and measure a force of up to 9000 N on the cell stack!
  Force adjustment possible throughout the entire experiment
- Built-in force, and temperature sensors!
  With optional gas pressure sensor and gas in- and outlet
- PAT-Solid-Core for easy assembly and reproducible results!
  Press and cycle solid-state batteries with 6 or 10 mm electrode diameter
- Cableless and highly sealed battery test cell!
  For precise long-term measurements of solid-state cell chemistries



Learn more on our product website: Scan me!

Download the data sheet (PDF): Scan me!

Or contact us directly:
+49 40 79012-734
sales@el-cell.com
www.el-cell.com

This content was downloaded from IP address 117.250.64.131 on 15/10/2026 at 10:30

Journal of The Electrochemical Society, 2022 **169** 020518

How do Depth of Discharge, C-rate and Calendar Age Affect Capacity Retention, Impedance Growth, the Electrodes, and the Electrolyte in Li-Ion Cells?

Roby Gauthier,[1,*] Aidan Luscombe,[1] Toby Bond,[1,2] Michael Bauer,[1] Michel Johnson,[1] Jessie Harlow,[1] A. J. Louli,[1,*] and J. R. Dahn[1,,z]**

[1]*Department of Physics and Atmospheric Science, Dalhousie University, Halifax B3H 4R2, Canada*
[2]*Canadian Light Source, Saskatoon S7N 2V3, Canada*

Lithium-ion cells testing under different state of charge ranges, C-rates and cycling temperature have different degrees of lithium inventory loss, impedance growth and active mass loss. Here, a large matrix of polycrystalline NMC622/natural graphite Li-ion pouch cells were tested with seven different state of charge ranges (0–25, 0–50, 0–75, 0–100, 75–100, 50–100 and 25%–100%), three different C-rates and at two temperatures. First, capacity fade was compared to a model developed by Deshpande and Bernardi. Second, after 2.5 years of cycling, detailed analysis by dV/dQ analysis, lithium-ion differential thermal analysis, volume expansion by Archimedes' principle, electrode stack growth, ultrasonic transmissivity and X-ray computed tomography were undertaken. These measurements enabled us to develop a complete picture of cell aging for these cells. This then led to an empirical model for cell capacity loss vs SOC range and calendar age. Although these particular cells exhibited substantial positive electrode active mass loss, this did not play a role in capacity retention because the cells were anode-limited during full discharge under all the tests carried out here. However, the positive electrode mass loss was strongly coupled to positive electrode swelling and electrolyte "unwetting" that would eventually cause dramatic failure.
© 2022 The Author(s). Published on behalf of The Electrochemical Society by IOP Publishing Limited. This is an open access article distributed under the terms of the Creative Commons Attribution 4.0 License (CC BY, http://creativecommons.org/licenses/by/4.0/), which permits unrestricted reuse of the work in any medium, provided the original work is properly cited. [DOI: 10.1149/1945-7111/ac4b82]

Manuscript submitted April 21, 2021; revised manuscript received December 14, 2021. Published February 4, 2022.

Supplementary material for this article is available online

Understanding and predicting the capacity fade of lithium-ion cells is still a huge challenge for researchers.[1] While it is generally understood that the primary cause of cell capacity fade at low C-rate is the growth of the negative electrode solid-electrolyte interface (SEI),[2–4] which leads to lithium inventory loss, for the general case it is still challenging to determine quantitatively how much each degradation mechanism (i.e. SEI growth, impedance growth, active mass loss, electrolyte oxidation, electrolyte degradation, electrode structural changes, etc.) contribute to capacity fade and how they evolve vs cycling time.[1,5]

Additionally, since we still don't clearly understand the physics and chemistry of many of these degradation mechanisms,[1,5] creating a mathematical model that accurately predicts capacity fade in the long-term with minimal experimental inputs is challenging. For example, many different mechanisms have been proposed to explain the continuous growth of the SEI on the graphite electrode after formation: diffusion of the electrolyte through nanosized SEI pores, electron tunneling, electron conduction or interstitial diffusion of neutral lithium through the SEI, "near-shore" SEI aggregation or SEI cracking.[6–8] However, researchers are still unable to determine which one is the major cause of SEI growth and as such, it is hard to be totally sure how the capacity loss will behave in advance. One of these SEI growth mechanisms, SEI cracking, was recently proposed by Deshpande and Bernardi.[8] By assuming that SEI cracking does occur, they developed a mathematical model for how capacity loss should evolve as a function of depth of discharge (DOD) and C-rate. According to Deshpande and Bernardi, SEI cracking can occur due to the expansion and contraction of graphite particles during charge and discharge. Note that while the depth of discharge (DOD) is generally defined as DOD = 100%—SOC, where SOC is the state of charge, in this work we define it as the difference between the upper cutoff SOC and lower cutoff SOC during cycling.

DOD is an important parameter that has a direct impact on cell degradation. Higher DOD generally results in a larger volume change of active particles during cycling, which may results in stress, cracking and cell degradation.[8,9] Definitions of the SOC and SOC range can be found on page 3 of the supplementary material.

The present work has two main goals:

1) Compare the model developed by Deshpande and Bernadi, which also includes a time-dependent capacity loss term,[3,8] to experiment and see if it can accurately predict capacity loss over time. If it does not, propose a new empirical model.
2) Study the degradations happening inside our cells as a function of SOC range, C-rate and temperature in order to gain new insights about the degradation modes that are occurring, which can be useful to propose new degradation models.

While the effect of SOC range on lifetime has been studied before,[8,10–23] there are few experimental studies on large SOC range matrices of cells that also include an extensive study of the changes which have occurred to the cells after years of testing. As such, point 2) can help researchers gain new knowledge about how the degradation of cells varies as the SOC range changes.

For this study, a large matrix of cells which consisted of two different cycling temperatures (20 °C and 40 °C), three different cycling rates (C/10, C/5, C/3), seven different SOC ranges (0%–-25%, 0%–50%, 0%–75%, 0%–100%, 75%–100%, 50%–100% and 25%–100%) was created, each condition having two cells, for a total of 96 cells. 3.0 V and 4.1 V were chosen as the lower cutoff voltage (LCV) and upper cutoff voltage (UCV) for the 0%–100% SOC range. 4.1 V was selected to limit oxidation at the positive electrode so that one can focus on the other mechanisms happening inside the cell. We think that understanding degradation at a UCV of 4.1 V is the first step before understanding degradation at higher voltage (e.g., 4.3 V). The cells consisted of polycrystalline NMC622/natural graphite (NMC622/NG) cells with VC211 (2% vinylene carbonate (VC) + 1% methylene methane disulfonate (MMDS) + 1% tris-trimethylsilyl phosphite (TTSPi)) as an additive blend in 1.2 M $LiPF_6$ EC:EMC 3:7 or 1.2 M $LiPF_6$ EC:DMC 3:7 electrolyte. The

*Electrochemical Society Student Member.
**Electrochemical Society Fellow.
[z]E-mail: jeff.dahn@dal.ca

VC211 additive blend was chosen because of its good performance in cells.[24] Natural graphite was used because of its continual swelling during extended charge-discharge cycling, which might or might not be a result of SEI cracking as proposed by Deshpande and Bernardi.[8,25] Polycrystalline NMC622 was used to study the effect of SOC range on the microcracking of the positive electrode. As a result, the cells used in this work are not intended to be as good as the "million-mile battery" of Harlow et al.,[26] instead they are intended to help researchers develop a deeper understanding of the degradation of lithium-ion cells.

dV/dQ analysis,[27] thickness and volume measurements,[28,29] X-ray computed tomography (CT),[30] ultrasonic transmission mapping (UTM),[31] and differential thermal analysis (DTA)[32] were used to gain new information about the degradation happening to the electrodes and electrolyte of our cells after long-term cycling. dV/dQ analysis was used to determine how the lithium inventory loss and the active mass losses change as a function of depth of discharge, C-rate and temperature.[27] The thickness and volume measurements were used to determine by how much the jelly roll thickness increased during degradation[28,29,33] and how much gas was being produced during cycling, X-ray CT was used to determine which parts of the cell were responsible for the thickness increase. UTM was used to determine if any electrolyte "unwetting", that can result in impedance growth, was occurring inside the cell during cycling,[31] and DTA was used to determine if changes to the electrolyte were occurring in our cells.[32]

Let us now makes some hypotheses about what we expect to find in the current study, first starting with goal 1). According to Deshpande and Bernardi,[8] cell capacity, Q, should evolve vs time t and cycle number, n, according to:

$$Q(t) = Q_0(1 - BDOD^2 n - K\sqrt{t}), \quad [1]$$

where Q_0, B and K are fitting constants, and DOD is the depth of discharge written as a fraction. Note that the last term, $K\sqrt{t}$, describes calendar aging.[3] This means that K is independent of DOD. Since $t \approx \frac{2n\ DOD}{Crate}$, where Crate is the C-rate written as a fraction, then:

$$Q(t) = Q_o\left(1 - B\ DOD\frac{Crate\ t}{2} - K\sqrt{t}\right), \quad [2]$$

If we choose Q_o, K and B to be equal to $Q_o = 1$ (normalized capacity), $K = 0.0005\ h^{-0.5}$ and $B = 0.0001\ h^{-1}$, we get Fig. 1:

The main features to remember from Fig. 1 are that, according to the model, the capacity loss should occur almost linearly at C/3 and capacity loss should become worse as either the C-rate or DOD increase. According to the model, capacity loss also gets worse as the C-rate increases for cells that would cycle at low DOD.

For goal 2), while we also expect to make unexpected discoveries along the way, we want to test if the following hypotheses are true or not:

- We expect capacity fade to be worse at high temperature, C-rate and DOD. According to the model of Deshpande and Bernardi (DB model), capacity fade vs time should increase linearly with DOD and linearly with C-rate. If SEI cracking is indeed happening as proposed by Deshpande and Bernardi, we expect that the non-linear graphite volume change vs SOC relationship will have a direct impact on the capacity fade.[28]
- We expect lithium inventory loss to be the major cause of capacity loss at low C-rate.
- We expect ΔV (the absolute value of the difference between average charge and discharge voltages) to be larger at high C-rate due to Ohm's law.
- We expect active mass loss per unit time to be larger at high DOD and C-rate due to the larger active particle volume change at high DOD and the larger number of volume changes that occur per unit time at higher C-rate.
- Due to the larger volume change of the graphite lattice vs the NMC622 lattice, as will be shown later in this work, and the larger irreversible volume change of natural graphite,[25] we expect that most of the thickness increase in the cell stack will come from the negative electrode. This may also result in more active mass loss at the negative electrode.
- Since vinylene carbonate is present in the electrolyte and because the upper cutoff potential is limited to 4.1 V, we expect minimal change to the electrolyte during the cell lifetime.
- It is expected that the thickness increase of the jelly roll after years of testing increases as the SOC range selected for the cycling also increases.[33]

Additionally, we want to determine possible causes for any impedance growth that occurs in the cells after years of testing.

While the techniques used in this work can't directly and fully resolve the hypotheses mentioned above or directly prove or disprove the SEI cracking mechanism, it is hoped that the knowledge gained here can help researchers in their quests to do just that.

Experimental

Pouch cells.—$Li(Ni_{0.6}Mn_{0.2}Co_{0.2})O_2$ (NMC622)/graphite (gr) 402035-size pouch cells were received dry (i.e., with no electrolyte) from Li-Fun Technology (Zhuzhou, Hunan, China). This work utilized two different NMC622 positive electrode active materials: one with an Al_2O_3 coating (NMC622A) and the other with a

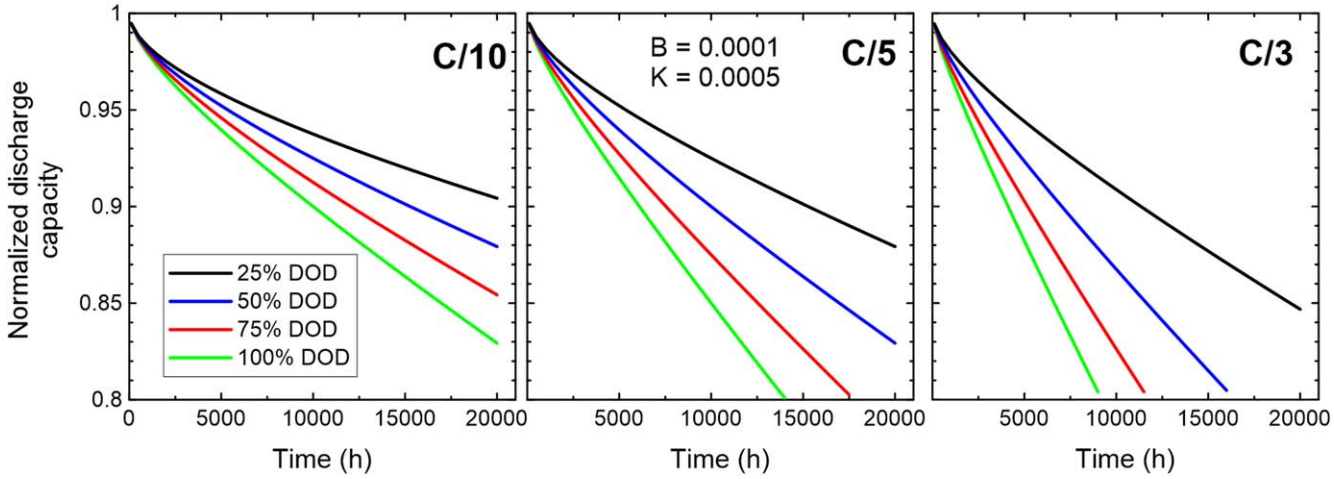


Figure 1. Deshpande-Bernardi model for the case of $Q_0 = 1$, $B = 0.0001\ h^{-1}$ and $K = 0.0005\ h^{-0.5}$.

proprietary coating (NMC622B).[34,35] The cell capacities at 4.3 V were ~250 mAh during formation for both cell types. After formation, the capacity at 4.1 V and C/10 was ~220 mAh. The NMC622 active material used in this work was composed of conventional secondary spherical agglomerations of smaller (hundreds of nm) primary particles. Before cells were filled with electrolyte, they were cut open in an argon-atmosphere environment and dried under vacuum at 80 °C for 14 h. Detailed information about the pouch cells can be found in Tables SI and SII.

Electrolyte preparation.—In most cases, the control electrolyte used was 1.2 M $LiPF_6$ in a 3:7 mixture by weight of EC (BASF, 99.95%, <20 ppm water) and EMC (BASF, 99.9%, <20 ppm water), respectively. However, for the cells that cycled at 20 °C, the control electrolyte used was 1.2 M $LiPF_6$ in a 3:7 mixture by weight of EC and DMC (BASF, <20 ppm water). VC211 ternary blend was used as the additive, which consists of 2 wt% VC (BASF, 99.5%, <100 ppm water), 1 wt% TTSPi (Tokyo Chemical Industry Co., Ltd. (TCI), > 95.0%) and 1 wt% MMDS (Guangzhou Tinci Co. Ltd., 98.70%).

The cells were filled with electrolyte in an argon-atmosphere environment and then vacuum-sealed at a gauge pressure of −90 kPa and a temperature of 165 °C using a compact vacuum sealer (MSK-115A, MTI Corp.). All electrochemical measurements reported in this work were performed on a minimum of two replicate pouch cells, to ensure experimental precision.

Formation.—Following filling with electrolyte, cells were maintained at 1.5 V and at room temperature for 24 h to allow time for the electrolyte to fully wet the electrodes. The cells were then moved into temperature boxes at 40.0 ± 0.1 °C and connected to a Maccor series 4000 battery tester for formation. First, cells were held at 1.5 V for 30 min to allow time for the temperature to stabilize and then they were charged at C/20 to 4.3 V. Cells were maintained at 4.3 V for one hour and then discharged at C/20 to 3.8 V. The electrolyte of each cell reduced during the first charge to form the initial SEI on the graphite at a cell voltage of 2.4 V and the cells produced ~0.3 ml of gas. Cells were degassed after formation and sealed again. After formation, the EIS Nyquist plot of the cells had a semi-circle diameter (R_{CT}) of ~40 $\Omega \cdot cm^2$ (see Fig. S1 (available online at stacks.iop.org/JES/169/020518/mmedia)).

Long term cycling test.—48 cells were placed in a temperature-controlled chamber at 40.0 ± 0.1 °C connected to a Neware battery cycler, while 48 cells were connected to a Neware battery cycler at room temperature (20 ± 2 °C). 24 of the cells at 40 °C were NMC622A/NG cells, while the rest were NMC622B/NG cells. The cells at room temperature were NMC622A/NG cells. The voltage ranges used for long-term cycling of each NMC622A/NG cell are summarized in Table I and the voltage ranges used for each NMC622B/NG cell are summarized in Table II. Each voltage range is associated with an approximate depth of discharge (DOD) or SOC range.

For each voltage range, three pairs of UC or LC cells were cycled at three different C-rates: C/10, C/5, and C/3, except for the cells at 20 °C. In the 20 °C case, three quartets of UC cells were cycled at these three different C-rates. Quartets were used instead of doublets since some data (100% SOC range checkup cycles) was missing before ~2000 h of testing for the first doublets that cycled for a total of 20000 h. As a result, a second doublet was cycled under the same conditions for ~7000 h. The data were then combined to determine the resulting capacity loss over the full 20000 h and to allow fitting. All NMC622A/NG cells that were not cycled at C/10 using a full voltage range (3.0 V–4.1 V) had checkup cycles every 500 h approximately, while all the equivalent NMC622B/NG cells had checkup cycles every 900 h approximately. The checkup cycles consisted of full range cycles (3.0 V–4.1 V) at C/10. Each main cycle and checkup cycle consisted of a CCCV charge and a CC discharge. A CCCV charge consists of a constant current (CC) charge from the lower cutoff voltage (LCV) to the upper cutoff voltage (UCV) followed by a constant voltage (CV) hold at the UCV until the current decreased to C/20. A CC discharge consists of a constant current (CC) discharge from UCV to LCV.

Cell thickness measurements.—The thickness of each pouch cell was measured in the center of the jelly roll region three times using a Mitutoyo linear gage. These three values were then averaged.

Cell volume measurements.—The change in volume of each cell was measured using Archimedes' principle. Cells after filling were suspended under an analytical balance (Shimadzu model AUW220D) by a metal wire and submerged in a beaker full of water and weighed. The process was repeated after formation to determine the volume of gas created during formation. The change in the volume of the cell was then calculated using the apparent change in mass due to the displacement of water, assuming the density of the water is ~1.00 g ml^{-1}. The cell weight underwater was measured again after degassing and yet again after long-term testing to determine the volume change during long-term testing.

UHPC measurements and manual dV/dQ analysis.—Ultra-high precision cycling (UHPC) was performed using the UHPC chargers at Dalhousie University after the cells went through approximately 20000 h of long-term cycling.[36] The NMC622/graphite cells were cycled between 3.0 V and 4.1 V at C/40 on the UHPC charger at 40 °C after an initial discharge from 3.8 V to 3.0 V. The first V(Q) charging curve of each relevant cell was used to perform a manual dV/dQ analysis which allows us to determine the Li inventory loss and the active mass of each electrode. This dV/dQ analysis was done using the software developed by H. M. Dahn et al.[27]

Automatic dV/dQ analysis on long-term cycling data.—Long-term Neware cycling data were automatically analyzed using a program written using the MATLAB R2020a software.[37] In the case of cells with checkup cycles, only the checkup cycles were analyzed and the remaining main cycles were discarded by the program. The program operates similarly to the software developed by H. M. Dahn et al., however instead of using a nonlinear least squares fitting algorithm (Levenberg-Marquardt method) to fit the calculated dV/dQ vs Q curve to the experimental dV/dQ vs Q curve, the program developed in this work calculated Chi-square from the difference between the theoretical and experimental dV/dQ vs Q curves for all negative electrode slippage, positive electrode slippage, negative active mass and positive active mass values within a discrete 4D matrix and selected the electrode mass and slippage values corresponding to the smallest Chi-squared. From now on, this method will be called the dV/dQ Matrix method. While the dV/dQ Matrix method is more computationally expensive than the Levenberg-Marquardt method, it gave the authors higher confidence in the results, since the dV/dQ Matrix method can help confirm that the

Table I. Voltage ranges applied to the NMC622A/NG cells. 18 cells are cycling for each voltage range. 12 cells at room temperature (20 °C) and six cells at 40 °C. This corresponds to 72 cells in total. These cells will be called fixed upper cutoff cells (UC cells).

	100% DOD 100%-0% SOC	~75% DOD 100%-25% SOC	~50% DOD 100%-50% SOC	~25% DOD 100%-75% SOC
Upper cutoff voltage	4.1 V	4.1 V	4.1 V	4.1 V
Lower cutoff voltage	3.0 V	3.495 V	3.6 V	3.77 V

Table II. Voltage ranges applied to the NMC622B/NG cells. Six cells are cycling for each voltage range, for a total of 24 cells. All are at 40 °C. These cells will be called fixed lower cutoff cells (LC cells).

	100% DOD 100%-0% SOC	∼75% DOD 75%-0% SOC	∼50% DOD 50%-0% SOC	∼25% DOD 25%-0% SOC
Upper cutoff voltage	4.1 V	3.777 V	3.649 V	3.527 V
Lower cutoff voltage	3.0 V	3.0 V	3.0 V	3.0 V

global minimum has been found, if the matrix range is large enough and the step size small enough. The automatic dV/dQ analysis was applied to the Neware data of a specific cell using the dV/dQ Matrix method on N equally spaced cycles from 0 h to ∼20000 h. The 4D matrix used for each cycle was a 150 × 75 × 10 × 5 matrix (positive electrode slippage precision x positive active mass precision x negative electrode slippage precision x negative active mass precision). When N = 6, the computational time for this matrix size was ∼10 min per cell. The positive electrode slippage was scanned from −0.5 mAh to 75 mAh, the positive active mass was scanned from 0.9 g to 1.64 g, the negative electrode slippage was scanned from 0 mAh to −10 mAh and the negative active mass was scanned from 0.98 g to 1.08 g. The scanning range was guided by the results of manual dV/dQ vs Q analysis.

Differential thermal analysis.—Lithium-ion differential thermal analysis (Li-ion DTA) is a non-destructive method of probing the state of the electrolyte inside a Li-ion cell. This method, first developed by Day et al., involves cooling the cell down to a cryogenic setpoint where the electrolyte in the cell is entirely frozen and tracking the temperature of the cell as it heats back to room temperature at a constant, controlled rate.[38]

In this case, the lower setpoint was −100.0 °C, while room temperature was 25 °C. This range was chosen to cover the entire freezing regime of the electrolyte. As well, this range is entirely within the liquid phase of the reference cell, which in this case is 1.0 M $LiPF_6$ in methyl acetate. This is beneficial, as the reference cell will act very similarly to a sample cell that does not go through phase changes in the regime of interest. In all cases, the cells were cooled at a rate of 3.0 °C min^{-1}, held at the low temperature for ten minutes to allow for electrolyte temperature equilibration, and heated at 1.0 °C min^{-1}. The direct temperature signals of the sample and reference cells can then be compared to determine the temperatures at which the sample cell underwent phase changes. As these phase changes are sensitive to electrolyte composition, we can identify changes to the electrolyte by comparing DTA curves.

Ultrasonic transmission mapping.—Ultrasonic transmission mapping was performed with an ultrasonic battery scanner (UBSC-LD50, Jiangsu Jitri-Hust Intelligent Equipment Technology Co., Ltd.).[31] A pair of ultrasonic focus transducers (2 MHz frequency, 30 mm focal distance, customized from Shantou Institute of Ultrasonic Instruments Co., Ltd.) were positioned on either side of the cell. Transducers and cells were immersed in low viscosity silicone oil which serves as an ultrasonic coupling agent. The transducers were installed on a 2-dimensional motion system with a precision of 0.2 mm to perform progressive scanning. The ultrasonic signal was emitted by a transducer on one side of the cell and received by the transducer on the other side. The driving source was a 200 V pulse signal with a pulse width of 250 ns, matching the 2 MHz ultrasonic transducer used here. The waveform of the transmitted signal was recorded with a collecting card. The peak-to-peak values of received transmission waves were converted into color heat maps to make pseudo color images. Cells were equilibrated at 3.8 V and degassed before the ultrasonic measurements were performed at 20 °C.

Computed tomography (CT) scans.—CT scans were collected at the Canadian Light Source synchrotron facility on the Biomedical Imaging and Therapy Insertion Device beamline (BMIT-ID).[39] This beamline provides a coherent, highly-collimated, monochromatic X-ray beam with parallel geometry, allowing for enhanced contrast through propagation-based phase imaging. The use of a monochromatic beam also minimizes the X-ray dose to the cell, as all incident photons contain sufficient energy to contribute to the acquired image. The high incident flux (3×10^{10} photons cm^{-2}·s) also allows for scans to be acquired many times faster than conventional CT, which limits artifacts from electrolyte motion. All scans were collected using a single-crystal LuAG ($Lu_3Al_5O_{12}$:Ce) scintillator paired with an optical camera. In this setup, the parallel X-ray projection image is optically magnified, as opposed to magnification using a conventional cone-beam source. This makes micron-level imaging of larger objects such as pouch cells possible, as very small sample-to-source distances are not required.

Two different setups were used to image at different scales. A lower-resolution, high-dynamic-range setup was used for imaging liquid electrolyte in the full cell, where a CCD camera (PCO 4000) with an effective pixel size of 8.9 μm was used with a monochromatic beam energy of 70 keV. Cells were scanned using 3000 projections (exposure time = 0.6 s) collected over a 180-degree range with the sample placed 40 cm from the detector. A higher-resolution setup was used to scan only the corner of the cells (for electrode thickness measurements) at a beam energy of 50 keV, where a high-magnification (5×) optical system (manufactured by Optique Peter) was used with an sCMOS camera (PCO Edge), providing an effective pixel size of 1.44 μm. 2500 projections were collected (exposure time = 1.0 s) with a sample-detector distance of 3 cm. All scans were reconstructed using the UFO-KIT software package using Paganin phase retrieval.[40] This technique processes the self-interference fringes present in synchrotron X-ray projections so that the contrast of the absorption image is enhanced using differences among the refractive indices of materials in the cell. After reconstruction, scans were manually aligned so that electrode planes were parallel to the image axes in the flat portions of the cell, with trilinear interpolation used for image resampling. Thickness measurements were taken manually by measuring the distance between surface planes at the high-contrast cathode/separator boundary.

Pressure measurements.—In-situ stack pressure measurements were performed using the apparatus detailed in Louli et al.[28] Pouch cells were volumetrically constrained within a rigid aluminum enclosure such that any volume expansion of the electrode stack would exert a force within the enclosure. The force within the enclosure was measured with subminiature load cells (LCKD-OMEGA Engineering) connected to DP25B-S-A (OMEGA Engineering) strain gauge panel meters. The load cells were fastened in the enclosure with the pouch cell, separated by a force distributing plate. The measured force was converted to kPa using the pouch cell area of 6 cm^2. For in situ measurements, the pouch cells were connected to a Neware battery testing system and cycled at C/20 between 3.0 V and 4.1 V. The analog 0–10 V output of the strain gauge panel meter was connected to an adjacent Neware channel to enable simultaneous electrochemical and pressure measurements from both the pouch cell and pressure sensor. These measurements were performed in a 40.0 ± 0.1 °C temperature box. Finally, the initial pressure at a state of charge (SOC) of 0% was subtracted from the data, so that the "zeroed" pressure at a SOC of 0% was 0 kPa.

Theory

dV/dQ analysis is a useful technique to deduce how much active mass loss and electrode slippage occur in a cell.[27] It is also possible to calculate lithium inventory loss, Q_{LIL}, from the information that dV/dQ analysis gives us. Figure 2 defines Q_{LIL} and also shows how to calculate it from the half-cell V(Q) curves, which are themselves linearly transformed using dV/dQ vs Q analysis. Figure 2 shows the V(Q) curve of the full-cell and the linearly transformed V(Q) curves of the anode half-cell and cathode half-cell before and after cycling. Figure 2 also helps to defines the variables Q_p, δ_n, Δ and Δ_0 from which the lithium inventory loss can be deduced; where $Q_p = Q_0\ (1 - m_p/m_{p0})$ is the capacity associated with positive active mass loss in mAh (Q_0 is the capacity of the cell during formation, m_p is the positive active mass and m_{p0} is the initial positive active mass before formation), δ_n is the absolute negative electrode slippage and δ_{n0} is δ_n before long-term cycling/after formation. $\Delta = \delta_n - Q_p$ is the relative slippage after long-term cycling, $\Delta_0 = \delta_{n0}$ is the relative slippage after formation. Note that if the cell is anode-limited, $\Delta > 0$ and if the cell is cathode-limited, $\Delta < 0$.

Note that the capacity of the full cell at top of charge (i.e. 4.1 V) has been fixed at 250 mAh in Fig. 2, the capacity of the cell during formation. With this convention, there is no positive electrode slippage, as each cell cycle is normalized to the same capacity at the top of charge. From Fig. 2, one concludes that the lithium inventory loss can be deduced from dV/dQ analysis using the following expression:

$$Q_{LIL} = (\Delta - \Delta_0) + Q_p = (\delta_n - \delta_{n0}), \qquad [3]$$

Next, since the notion of depth of discharge is closely related to active particle volume change, which is itself related to cracking and cell degradation, it is also important to have a good understanding of active material expansion during lithiation and to look at the change in pressure of a cycling constrained full cell.[8,41] Figure 3 shows that during charging of a full cell to 4.1 V, the graphite crystalline structure expands by 6%, while the NMC622 crystalline structure contracts by 2% due to the addition or removal of lithium atoms, respectively, to/from their intercalation structures.[28,42] The presence of a plateau can be seen in the 50%–90% SOC range of Fig. 3a which is a consequence of the 2 to 2 L transition of graphite.[43] The expansion and contraction of the graphite and NMC622 structures multiple times during cycling create stresses inside the active particles and on any surface layers which may result in cracking and cell degradation. Figure 3d shows that the *c*-axis of NMC622 expands during charging of the cell to 4.1 V, while Fig. 3c shows that the a-axis contracts, which is the reason for the overall contraction of the NMC622 lattice during charging in Fig. 3a, since the volume of a hexagonal unit cell is given by $a^2c\ \sin(60°)$.[42]

However, the overall volume change of the cells during extended cycling not only depends on the crystalline structure of the active materials but also on the possible cracking of primary and secondary particles, possible jelly roll deformation and SEI growth.[25,29,44] Figure 3b shows the change in pressure of NMC622A/NG and NMC622B/NG constrained full cells during cycling. One can see the 2 to 2 L transition of graphite in that figure. However, a slight slope is present which was not present in the average of the graphite and NMC622 fractional volume vs SOC in Fig. 3a. The additional contributions mentioned above might explain this. Note that the average of the graphite and NMC622 fractional volume in Fig. 3a, the black curve, considers the initial relative volume ratio of the negative and positive electrode (1.8:1). Also, note that gas will be forced to the gas bag and will not be registered in the pressure measurements that were done here.

Results

Experimental cycling data.—To accomplish the two goals mentioned in the introduction, 96 polycrystalline NMC622/NG cells were first formed on a Maccor charger. After our cells went through their formation protocol, cells were degassed and then connected to a Neware charger and tested at room temperature or in temperature boxes at 40 °C. After the cells went through an adequate amount of cycles, the exact DOD value of each cell was calculated and was found to be slightly different than the approximate values predicted in Tables I and II. Figure 4 shows these exact values. For a specific approximate DOD (e.g. ∼75% DOD) one can see that the exact DOD values as a function of C-rate do not change more than ± 5%. Each plateau in Fig. 4 shows the data points of cells at C/10 (red squares), at C/5 (blue circles) and at C/3 (green triangles),

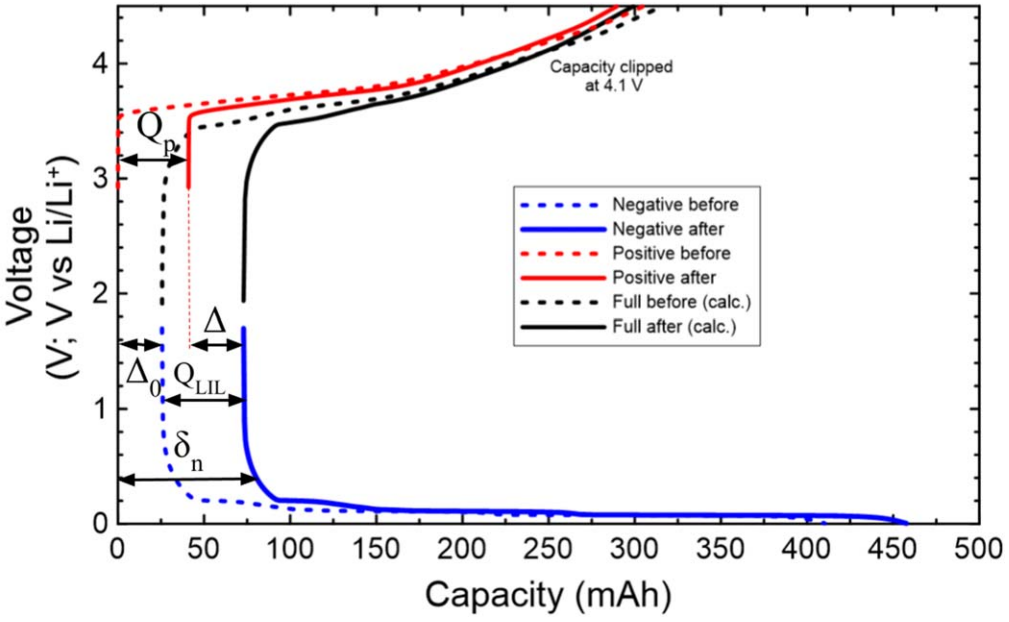


Figure 2. Figure used to define the variables Δ, Δ_0, Q_p, Q_{LIL} and δ_n for a specific set of positive and negative electrode half-cell V(Q) curves. In the present figure, natural graphite and NMC622 reference half-cell V(Q) curves were used and linearly transformed using initial and final slippages of $\Delta_0 = 26$ mAh and $\Delta = 73$ mAh, initial and final active positive mass of $m_{p0} = 1.40$ g and $m_p = 1.24$ g and equal initial and final active negative mass of $m_{n0} = m_n = 1$ g. "Before" could represent a cell immediately after formation and "after" could represent a cell after long-term cycling. The numbers used here were selected as examples based on later results. The full-cell V(Q) curves shown were then calculated by subtracting the negative half-cell voltages from the positive half-cell voltages.

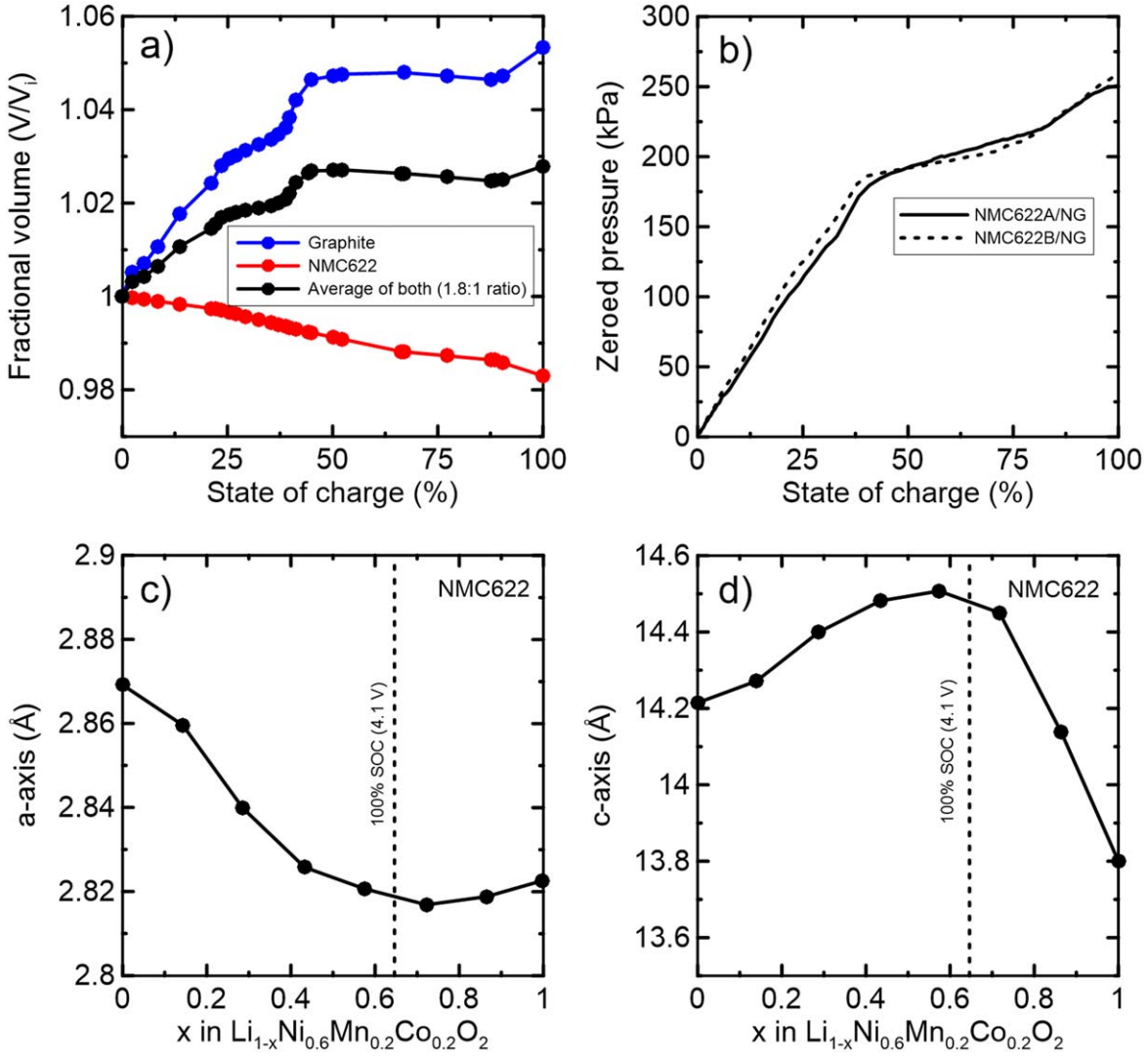


Figure 3. (a) Fractional volume change of the unit cell of graphite, NMC622 and the average of both as a function of state of charge, (b) zeroed pressure as a function of state of charge of NMC622A/NG and NMC622B/NG cells, (c) a-axis length and (d) *c*-axis length of the unit cell of NMC622 as a function of delithiation. Note that the average of the graphite and NMC622 fractional volumes in (a) consider their initial relative volume ratio (1.8:1). The full state of charge at 4.1 V is indicated for (c) and d). For a) and (b), a state of charge of 100% is the state of charge that occurs when the cell is at 4.1 V. NMC622 data in (a), (c) and (d) were digitized from Lee et al.[42] and graphite data in (a) were digitized from Louli et al.[28]

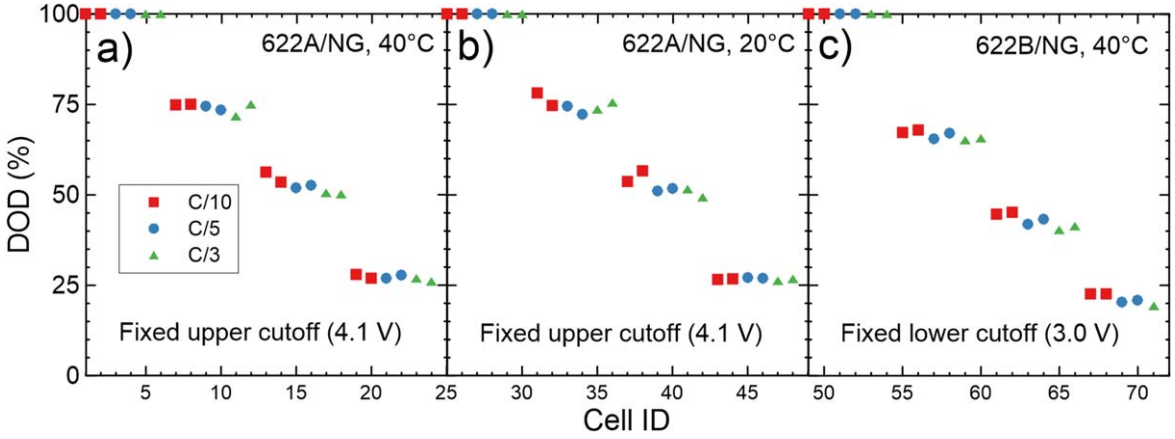


Figure 4. Exact initial DOD values for UC and LC cells at t ≈ 0 h. For each plateau, the first two data points correspond to cells cycling at C/10 (red squares), the two in the middle to cells cycling at C/5 (blue circles) and the last two to cells cycling at C/3 (green triangles). Each DOD value is calculated using the full-range (3.0 V–4.1 V) cells at the same rate as a 100% DOD reference. Figure 4(a) shows the initial DOD values for NMC622A/NG UC cells that cycled at 40 °C, Figure 4(b) shows data for some of the NMC622A/NG UC cells that cycled at 20 °C and Figure 4(c) shows the NMC622B/NG LC cells that cycled at 40 °C.

respectively from left to right. The LC cells have a smaller DOD than the UC cells by approximately 5%, except for the 100% DOD cells. Furthermore, all cells that permanently cycled in the 3.0 V–4.1 V range had their DOD values set to 100% DOD for all C-rates and all cells used the 100% DOD cells at the same C-rate as a reference to calculate their DOD values. The exact DOD values from Fig. 4 are used in this work when needed.

Each cell was cycled for approximately 20000 h using the appropriate protocols, which are mentioned in the Experimental Methods section. As an example, Fig. 5 shows the discharge capacity vs time of the fixed lower cutoff (LC) cells that cycled at 40 °C and C/3 for the four different voltage/SOC ranges mentioned in Table II. In this figure, one can see both the data of the main cycles (C/3 cycles at a DOD lower or equal to 100% with initial capacities of ∼50 mAh, ∼75 mAh, ∼150 mAh and ∼210 mAh. Those are the most frequent cycles.) and the data of the checkup cycles (100% DOD checkup cycles at C/10 with initial capacities of ∼210 mAh every ∼900 h). The approximate number of cycles after 20000 h of testing for all four SOC ranges are also shown. The depth of discharges that corresponds to each color are stated in the caption. Starting with the main cycles, one can see that the capacity loss decreases as the DOD decreases. Surprisingly, at ∼25% DOD the capacity vs time data of the main cycles is virtually flat during the whole ∼20000 h of cycling although these cells went through 16680 short cycles in total, compared to 3370 full cycles at 100% DOD. This is attributable to the shorter ∼25% DOD cycles that allow approximately four times more cycles in the same amount of time relative to the 100% DOD cycles (because capacity is proportional to time at constant current). Additionally, the flat capacity at ∼25% DOD can be explained by having a close look at how the capacity of the full-cell curve is changing in Fig. 2 between 3.0 V and 3.5 V vs between 3.0 V and 4.1 V as the negative electrode slips to the right. The ∼50% DOD and ∼75% DOD data can be explained similarly. For the curious reader, the discharge capacity data for the other 88 cells can be found in Figs. S3 to S5.

Figure 6c shows an expanded view of the checkup cycles from Fig. 5 around ∼210 mAh. The data from the four different SOC ranges are shown in different colors and each corresponds to a specific DOD as mentioned in the caption. Similar expanded views are also shown for the C/10 and C/5 data in Figs. 6a and 6b, again for the LC cells that cycled at 40 °C. Figure 6 shows that capacity loss is worse at high DOD. The capacity loss also became worse at a high C-rate, but only for the ∼75% DOD and 100% DOD cells. For the ∼25% DOD and ∼50% DOD cells, the capacity loss is virtually constant vs C-rate. We will later compare the DB model to the experimental data shown in Fig. 6 to see if they agree with each other.

Figure 7 shows the normalized capacity loss between 0 h and 20000 h for all the cells from this work. Figures 7a and 7b show that capacity losses are similar between UC and LC cells cycling at 40 °C and Fig. 7 shows that the data is linear vs DOD. However, Fig. 7c shows that capacity loss for cells cycling at 20 °C is non-linear as a function of DOD. This could be caused by a mix of contributions from diffusion to/from the overhang/overlap and SEI growth or other unknown phenomena (see Fig. S6).[45] As mentioned in the caption of the figure, since the UC cells that cycled at 40 °C were missing their checkup cycles for the first ∼2000 h of cycling, two different types of extrapolations (linear and square root extrapolations) were used to approximate the overall capacity loss during the full 20000 h of cycling. Similarly, the UC cells that cycled at 20 °C were also missing their checkup cycles in the first ∼2000 h of cycling, so new cells were put on Neware chargers to collect new data. This data was then adequately combined with the old data to get the results in Fig. 7c. Figure S6 shows the combined experimental data for the cells that cycled at C/10. The curious reader can also find the average voltage of the fixed UC cells that cycled at 20 °C in Fig. S14.

Comparison of the experimental cycling data to theoretical models.—Next, the experimental data in Fig. 6a, the LC cells data at C/10, was fitted to the DB model, as shown in Fig. 8. Additionally, the more traditional square root of time model to describe capacity vs time, Q(t), was also used:[3,46,47]

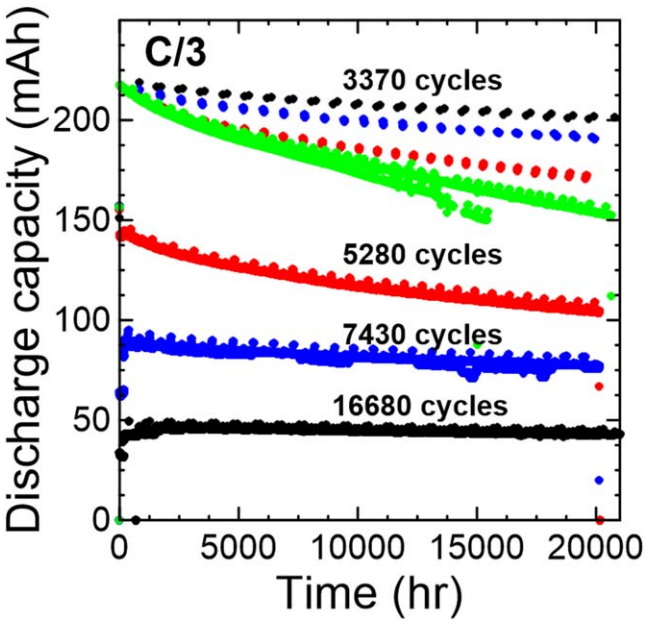


Figure 5. Discharge capacity vs time of the LC cells that cycled at 40 °C and C/3. The cycling data for 100% DOD (green data), 75% DOD (red data), 50% DOD (blue data) and 25% DOD (black data) are shown from top to bottom. The cycle numbers of each at 20000 h are also shown. The black, blue, and red data points near the top of the panel are the data for the 100% DOD C/10 checkup cycles.

$$Q(t) = Q_0(1 - A\sqrt{t}), \quad [4]$$

where Q_0 and A are fitting parameters, and t is time. Equation 4 can be derived by assuming that the growth of the solid electrolyte interface (SEI), which results in lithium inventory loss, is inversely proportional to the thickness of the SEI. In other words, the model assumes that SEI growth is diffusion-limited but in reality things are much more complicated.[48] Inspired by the linear trend in Figs. 7a and 7b, Eq. 4 can be rewritten as the empirical model that follows:[3]

$$Q(t) = Q_0(1 - (A_0 DOD + B_0) \sqrt{t}), \quad [5]$$

where A_0 and B_0 are fitting constants, and DOD is the depth of discharge written as a fraction. Note that Eqs. 4 and 5 assume that our cells are anode-limited and this is indeed confirmed later in this work.

Figure 8 shows the results of fitting both the DB model and the square root of time model to the data of LC cells that cycled at C/10. One can clearly see that the new model proposed, Eq. 5, is fitting and follows the trends of the experimental data better than the DB model in all cases. Figure 8 suggests that the linear contribution in Eq. 2 is the cause of the worse fit. Note that the fits to both models were done by minimizing the sum of the χ^2 of all four curves in Figs. 8a–8d as a whole, instead of minimizing each curve individually.

To get further evidence that the DB model doesn't appropriately describe our experimental data, the values of the fit obtained for Fig. 8 were used to predict capacity vs time at C/5 and C/3. The results are shown in Fig. 9. One can clearly see that capacity loss is overestimated for cells that cycled at low DOD and high C-rate. Additionally, the DB model is much more linear vs time than the experimental data. Note that while these results show that the DB model is not describing the data well in its current form, these results can't confirm if SEI cracking is occurring or not at the negative electrode in our cells.

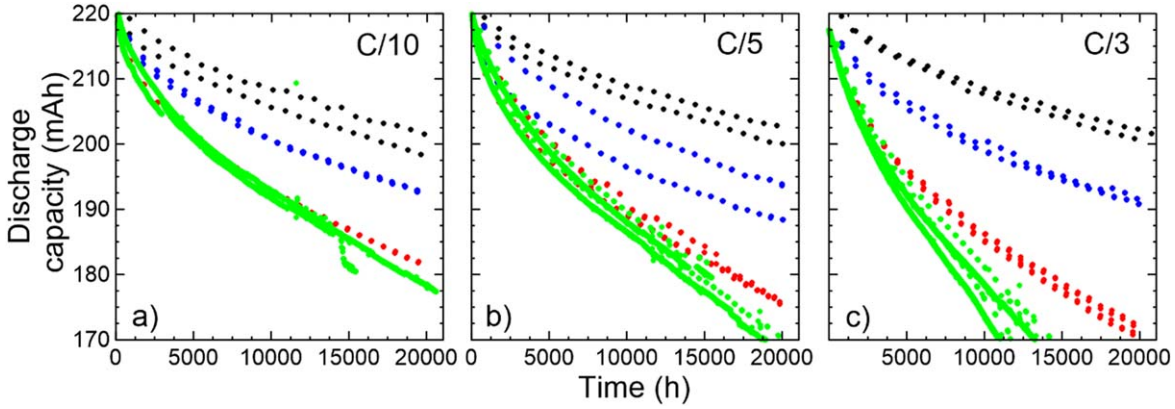


Figure 6. Discharge capacity vs time of the checkup cycles for the LC cells that cycled at 40 °C and (a) C/10, (b) C/5 and (c) C/3. The cycling data for 100% DOD (green data), ∼75% DOD (red data), ∼50% DOD (blue data) and ∼25% DOD (black data) are shown from bottom to top.

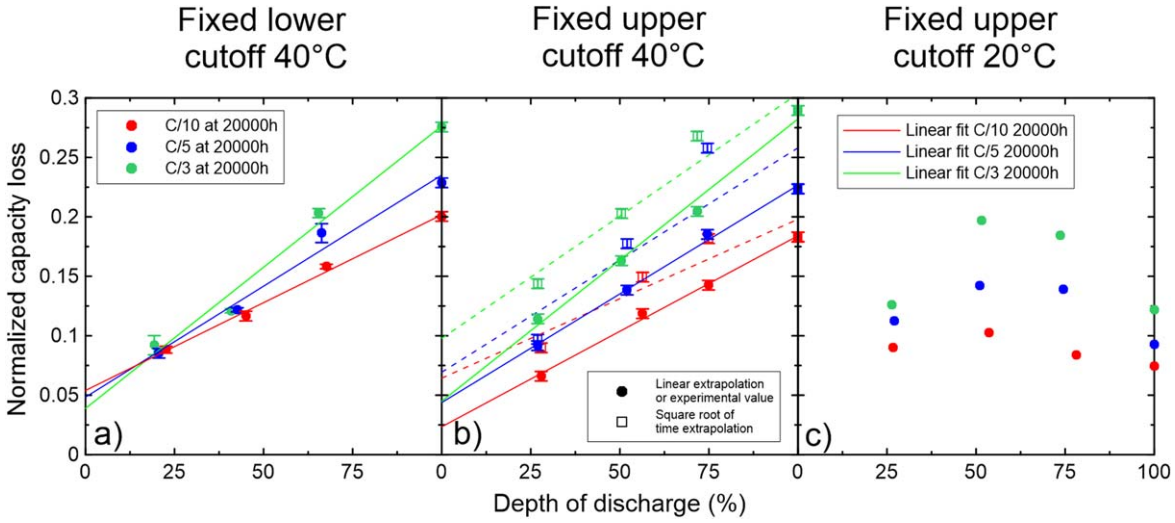


Figure 7. Normalized capacity loss during long-term cycling between 0 h and 20000 h of (a) LC cells that cycled at 40 °C, (b) UC cells that cycled at 40 °C and of (c) UC cells that cycled at 20 °C. Checkup cycles are used here to calculate the capacity loss. Since UC cells are missing their initial checkup cycles, linear extrapolation (solid circle data and solid lines), or square root of time model extrapolation,[46,47] as justified later in the text, (open square data points and dashed lines) to 0 h was used to estimate the capacity loss of the UC cells that cycled at 40 °C and a combination with new data was done for the UC cells that cycled at 20 °C (see Fig. S6 for an example at C/10). Since no data were missing for LC cells, no extrapolation was needed for the data in a). Red, blue, and green data are used to show the results from different C-rates (C/10, C/5, and C/3 in order). Note that no extrapolations were needed for the 100% DOD UC cells that cycled at 40 °C and as such, the normalized capacity loss values in (b) are the same for both the linear and square root of time extrapolations.

From Figs. 8 and 9, it looks like a new model is needed to describe the data. While Eq. 5 is doing a great job at fitting the data at C/10 in Fig. 8, the model still doesn't take into account the effects of impedance growth on capacity loss, which are no longer negligible at C/5 and C/3. As such, understanding the effects of impedance on capacity loss is the next step before going further.

Contribution of impedance growth to capacity loss.—While the square root of time model shown in Eqs. 4 and 5 considers the growth of the SEI, it does not consider the contribution of impedance growth on capacity loss.[49] To solve this problem, a first-order impedance term, as justified later in the text, can be added to Eq. 4. Additionally, a third term, $Q_{CV}(t)$, is added to consider the impact of a constant voltage (CV) charging step at 4.1 V, which results in an impedance-related capacity gain. The following approximation is proposed for the discharge capacity vs time:

$$Q_d(t) = Q_0(1 - A\sqrt{t}) - \frac{1}{2}\left(\frac{dQ}{dV}\bigg|_L + \frac{dQ}{dV}\bigg|_U\right)(\Delta V(t) - \Delta V_0) + Q_{CV}(t), \quad [6]$$

where $\frac{dQ}{dV}\big|_L$ is the absolute value of the derivative of the capacity with respect to the voltage at the LCV during discharge, $\frac{dQ}{dV}\big|_U$ is the absolute value of the derivative of the capacity with respect to the voltage at the UCV during charge. $\Delta V(t)$ is the absolute value of the difference between the average charge voltage and the average discharge voltage at time t and ΔV_0 is $\Delta V(t)$ at time $t = 0$. The variable ΔV is closely related to the impedance of the cell by Ohm's law ($\Delta V \approx 2RI$). The capacity gain that occurs during the constant voltage step, $Q_{CV}(t)$, can be described by an empirical linear term, as justified by Fig. 10.

The linear impedance term in Eq. 5 can be explained using Fig. 11. Figures 11a and 11b show the effect of impedance growth on discharge V(Q) curves for two different LCV: 3.0 V and 3.77 V. In this example, exaggerated for clarity, we show three discharge V (Q) curves in both Figs. 1a and 1b: The upper V(Q) curves are without impedance effects, the second curves have a 0.2 V ΔV

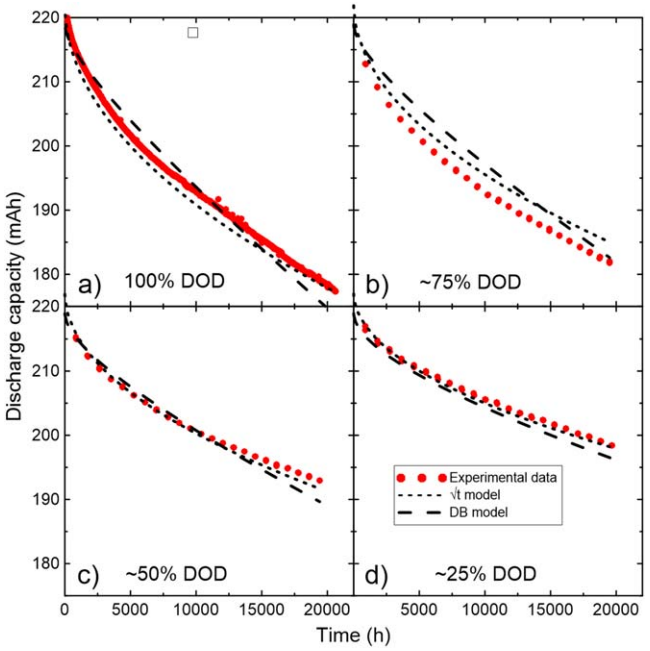


Figure 8. Fit of the DB model, Eq. 2, and of the square root of time model, Eq. 5, to the experimental data of LC cells that cycled at 40 °C and C/10. For the DB model, the results of the fit were $Q_o = 219$; $B = 0.00013$; $K = 0.000508$. For the square root of time model, the results of the fit were $Q_o = 222.1$; $A_0 = 0.00085$; $B_0 = 0.000558$. Note that since $K\sqrt{t}$ in Eq. 1 describes calendar aging in the DB model, K was not allowed to vary vs DOD.

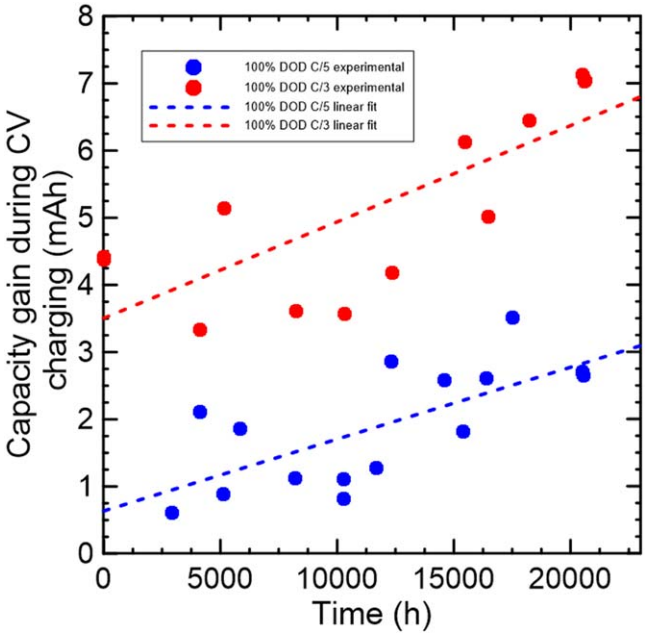


Figure 10. Capacity gain that occurred during the constant voltage (CV) step of the cycling of the 100% DOD cells from the LC cell batch at C/5 and C/3 at 40 °C as a function of cycling time. The experimental data are fitted with a linear model. The fitting results are $0.0001070\ t + 0.6350$ for C/5 and $0.0001434\ t + 3.505$ for C/3, where t is the cycling time in hours.

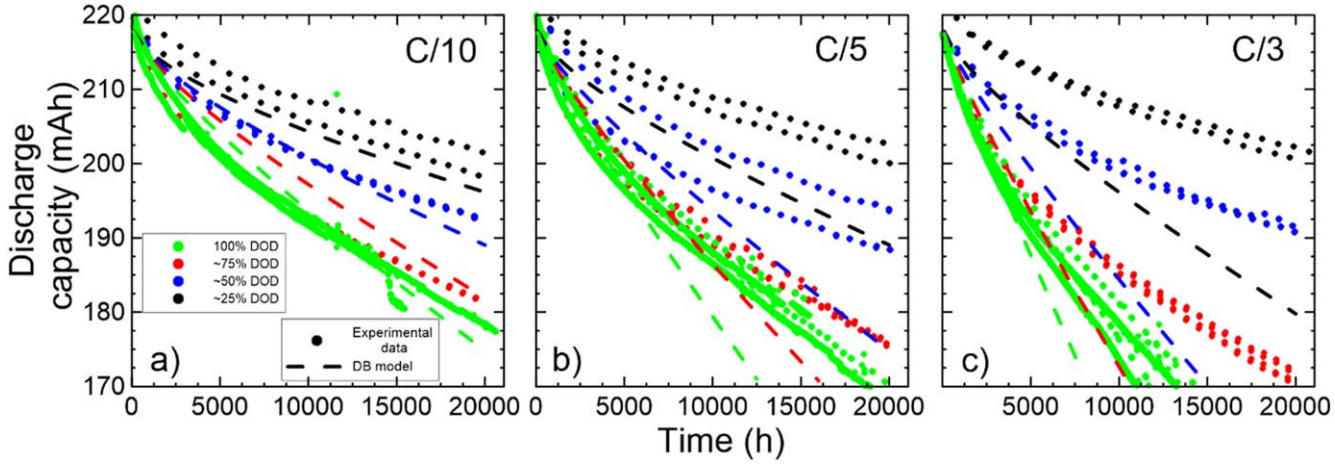


Figure 9. Comparison of the DB model predictions (dashed lines) and experimental data (dots and solid lines) for LC cells that cycled at 40 °C. The parameters used in the DB model for Fig. 9 are those that were determined from the fits in Fig. 8.

growth, and the lowest curves have a 0.4 V ΔV growth. Figure 11 shows that, since the slope of the V(Q) curve is different at different voltages, a cell that is discharged to 3.77 V without a constant voltage step will have a much higher impedance-related capacity loss than if the cell is discharged all the way down to 3.0 V. As a first-order approximation, the resulting capacity loss is $\frac{1}{2}\frac{dQ}{dV}\Big|_L (\Delta V(t) - \Delta V_0)$ during discharge. Similar reasoning can be applied to the constant current part of the V(Q) curve during charge, which results in a capacity loss of $\frac{1}{2}\ \frac{dQ}{dV}\Big|_U (\Delta V(t) - \Delta V_0)$. The $Q_{CV}(t)$ term is necessary to add in the case of CCCV charging (constant current followed by constant voltage) but is unnecessary in the case of CC charging (constant current only).

Figure 11c shows $\frac{dQ}{dV}$ as a function of voltage during charge and discharge of the first cycle of a NMC622A/NG cell. Remember that all $\frac{dQ}{dV}$ values in this work are absolute values. A voltage shift between the charge and discharge $\frac{dQ}{dV}$ vs V curves can be seen and this is caused by cell impedance. We can also see that a LCV of 3.0 V would result in a much lower impedance-related capacity loss than a LCV of 3.4 V (around 18 times less). Note that impedance-related capacity loss can be recovered by switching to a lower cycling current I, switching to other voltage cutoffs with small dQ/dV values or lowering the cutoff current during the end of CCCV charging.

Two assumptions have been made to write Eq. 6 since we only include the first order impedance term of a Taylor series, and we assume that the impedance ΔV can be treated as a resistor. To first

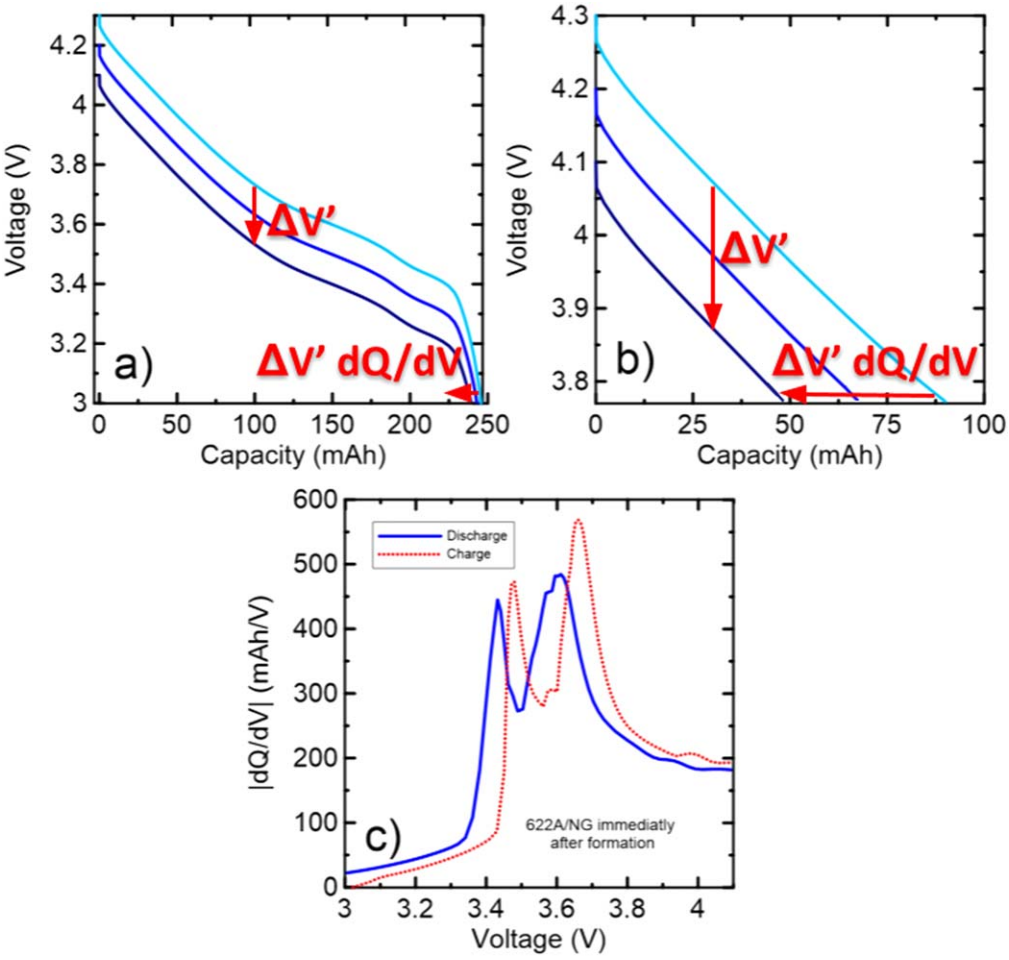


Figure 11. Figures 1a and 1b show the effect of impedance growth on discharge V(Q) curves for two different LCV: 3.0 V and 3.77 V. Three curves in each upper panel are shown. The upper lighter blue ones show V(Q) curves with no ΔV growth, the middles blue ones show V(Q) curves with a ΔV growth of 0.2 V and the bottom and darker blue ones show V(Q) curves with a ΔV growth of 0.4 V. In the figure, $\Delta V' = ½ (\Delta V - \Delta V_0)$. The slope of the V(Q) curve at the LCV has an impact on capacity loss. Figure 1c shows the derivative of the capacity with respect to voltage during charge and discharge of NMC622A/NG cells during the first cycle after formation measured at C/20 and at 40°C. The shift between the two curves is mostly due to the non-zero value of ΔV at $t = 0$ which results in a ~0.1 V voltage shift.

order, these assumptions are correct, however more complete models may be needed in some specific cases,[50–52] even more if the user can't afford the amount of error that a first-order approach offers. For example, if the impedance growth is large and/or $\frac{d^2Q}{dV^2}\Big|_L$ or $\frac{d^2Q}{dV^2}\Big|_U$ are large and/or non-ohmic effects are dominant, the second and possibly higher-order terms may need to be included. One can find a more detailed model and further explanations after Fig. S6 in the supplementary information.

Understanding trends in impedance growth is important since it directly impacts capacity loss as shown in Fig. 11. To better understand how impedance growth depends on the SOC range and the C-rate, the $\Delta V(t)$ data of checkup cycles for the UC and LC cells that cycled at 40 °C was fitted using a linear model of the form $\Delta V(t) = C\,t + D$. This linear model was justified by the very linear nature of the experimental data (see Fig. S8). The values of C obtained from those fits are shown in Figs. 12a and 12b. Since the value of D was virtually constant for all fits, only the average values and the standard deviations for parameter D are shown. Graphs that show a comparison of those fits to the experimental data for the case of the LC cells that cycled at 40 °C can be found in Fig. S8. Figures 12a and 12b show that parameter C grows approximately quadratically as a function of DOD, that parameter C is virtually the same for UC and LC cells at C/10, but that the symmetry between UC and LC cells is broken at a high C-rate. This suggests that damage to the cells that cause impedance growth is worse at high C-rate, high average voltage, and high DOD.

Figures 12c and 12d show the impedance contribution to the capacity loss for the checkup cycles of each LC and UC cell by using only the impedance term in Eq. 6. One can see that the impedance contribution to the capacity loss for the UC cell that cycled at C/3 and 100% DOD is ~3.8 mAh after 20000 h of cycling, while the impedance contribution to the capacity loss for the UC cell that cycled at C/10 and 25% DOD is ~0.25 mAh.

Figure 13a shows the same data as Fig. 12a vs C-rate instead. One can clearly see the linearity of parameter C vs C-rate and this suggests that impedance growth of the checkup cycles for UC cells follows Ohm's law at 40 °C. The results of the linear fit are also shown in Fig. 13a for the curious reader.

Figure 13b shows the results of linearly fitting $\Delta V(t)$ with $Ct + D$ for both the C/10 checkup cycles (blue data) and the main (C/10, C/5, and C/3) cycles (red data) of UC cells that cycled at 100% DOD and 40 °C. When dividing parameter C for the main cycles by parameter C for the checkup cycles, we find a factor of 1.15 at C/5 and a factor of 1.75 at C/3 (and a factor of 1 at C/10, since the checkup cycles and main cycles are equivalent). This is unexpected since one would have expected a factor of 10/5 = 2 at C/5 and a factor of 10/3 = 3.33 at C/3 from Ohm's law. As such, one can't use Ohm's law to calculate the impedance growth of the main cycles

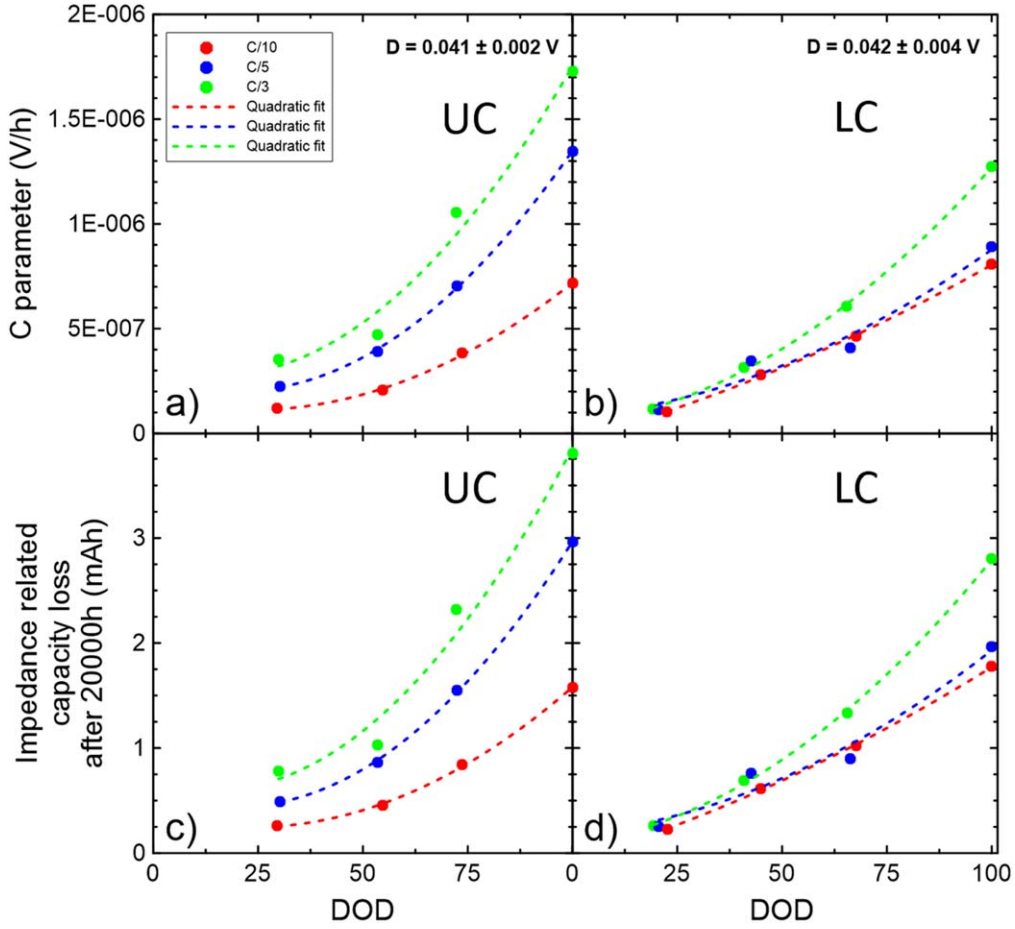


Figure 12. (a) and (b) show parameter C vs DOD, in percent, as calculated from fitting the experimental $\Delta V(t)$ data of the checkup cycles of UC and LC cells that cycled at 40 °C using the linear model $\Delta V(t) = C\ t + D$. (c) and (d) show the impedance related capacity loss after 20000 h of long-term cycling for the same cells as calculated from panels (a) and (b) and the impedance term in Eq. 6. The capacity loss shown in panels (c) and (d) is only applicable to the checkup cycles. Quadratic fits were applied to the C vs DOD data and to the impedance-related capacity loss vs DOD data. The result of the C/10 fit for panel (a) is $1.026 \times 10^{-10} \times^2$–$4.73 \times 10^{-9} \times + 1.658 \times 10^{-7}$ and $2.8839 \times 10^{-11} \times^2 + 5.50 \times 10^{-9} \times - 3.32 \times 10^{-8}$ for the C/10 fit in (b), where x is the number on the *x*-axis in Figs. 12a and 12b and represents the depth of discharge (DOD). In all panels, C/10 data is shown in red, C/5 data is shown in blue and C/3 data is shown in green.

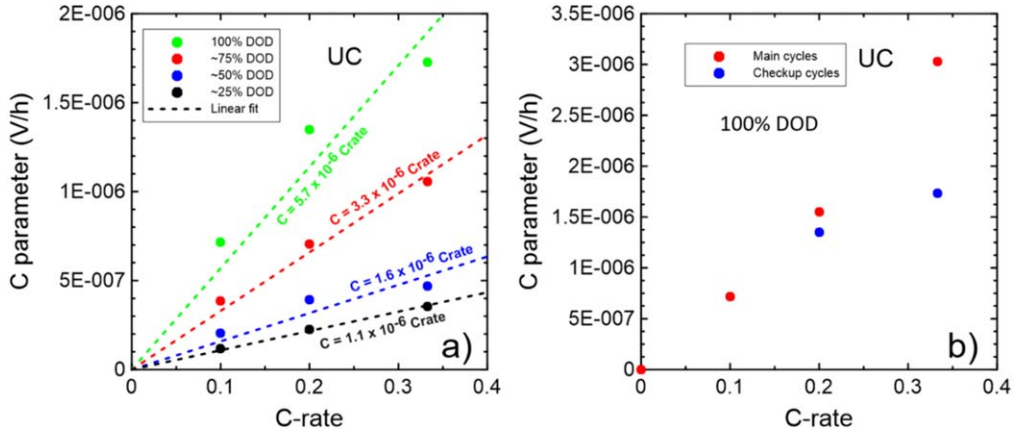


Figure 13. (a) Parameter C vs C-rate as calculated from fitting the experimental $\Delta V(t)$ data of the checkup cycles of UC cells that cycled at 40 °C using the linear model $\Delta V(t) = C\ t + D$. Different colors represent different DOD as shown in the legend. Linear regression results of C vs C-rate are also shown. (b) Parameter C vs C-rate as calculated from fitting the experimental $\Delta V(t)$ data of the C/10 checkup cycles (blue) and main (C/10, C/5 and C/3) cycles (red) of UC cells that cycled at 100% DOD and 40 °C using the linear model $\Delta V(t) = C\ t + D$.

from the impedance growth of the checkup cycles. For the curious reader, Figs. S10 to S12 show examples of linear fits to ΔV values of main cycles for cells that cycled at C/10 and Fig. S13 shows a summary of parameters C and D obtained by fitting ΔV values of main cycles at C/10, C/5 and C/3 for all cells.

For the case of UC cells that cycled at 20 °C, impedance growth was much smaller, as shown in Fig. S5. The largest ΔV growth was

approximately ∼0.02 V at C/3 and 100% DOD, which corresponds to a capacity loss after 20000 h of ∼2.3 mAh. At C/5 and 100% DOD, the ΔV growth was half of this, corresponding to a capacity loss of ∼1.2 mAh after 20000 h. For cells that cycled at lower DOD or at C/10, impedance growth was virtually nonexistent.

Square root of time model combined with an impedance term.—Now that we introduced the effect of impedance on capacity loss and measured how impedance changes when the cells cycle under different conditions, we can start using Eqs. 4 and 6 to fit the capacity vs time of all our cells. Figures 14a and 14b show the results of these fits, more specifically parameter A from Eqs. 4 and 6, for LC and UC cells that cycled at 40 °C. We can clearly see that parameter A is indeed mostly linear vs DOD. To show the goodness of fit for some of our cells, Fig. 14d shows a comparison between the experimental data and the fits for LC cells that cycled at 40 °C and C/10. Similarly, Fig. S15 shows the same comparison for UC cells that cycled at 40 °C and C/10. The fits in Figs. 14d and S15 are very good.

Do not confuse the fitting results in Figs. 8 and 14c. In Fig. 8, the fit was done, using Eq. 5, by minimizing the sum of Chi-Squared between the fit and the experimental data from all four cells as a whole, while in Fig. 14c, the fits were done individually on each cell by using Eqs. 4 or 6, depending on the C-rate.

Can the models developed in this work predict capacity loss vs time? Figure 15 shows that it is possible to quantitatively predict the discharge capacity vs time data of some of our cells that cycled at C/10 and to qualitatively predict the discharge capacity vs time data of some of our cells that cycled at C/5 and C/3. To make the predictions, the first 2000, 5000 or 6000 h of cycling were fitted with a pure lithium inventory loss model (i.e., Eq. 4) or by combining the lithium inventory loss term with an impedance term (i.e., Eq. 6) and then projecting the capacity vs time for the full 20000 h of data. Interestingly, since ΔV growth is small at low DOD and C-rate, as it was shown in Figs. 12 and 13, one can predict the data for the 100% DOD and ∼25% DOD cells at C/10 without any impedance term and very little data, as seen in Figs. 15a and 15b. Figures 15c and 15d show that adding an impedance term can explain most of the difference between the square root of time model (red dashed lines) and the experimental data (blue lines) for cells tested at C/5 and C/3. The predictions that used the impedance term (i.e., Eq. 6) are shown using green dashed lines or solid red lines. For the case of the green dashed line fit, the first 6000 h of ΔV(t) data were fitted using a linear model, while for the case of the solid red lines, the full 20000 h of ΔV(t) data were fitted using a quadratic model. Table III shows the parameters that were obtained when fitting the experimental capacity vs time data in Fig. 15. Note that the Q_{CV} term in Eq. 6 was calculated from the linear fit in Fig. 10 for

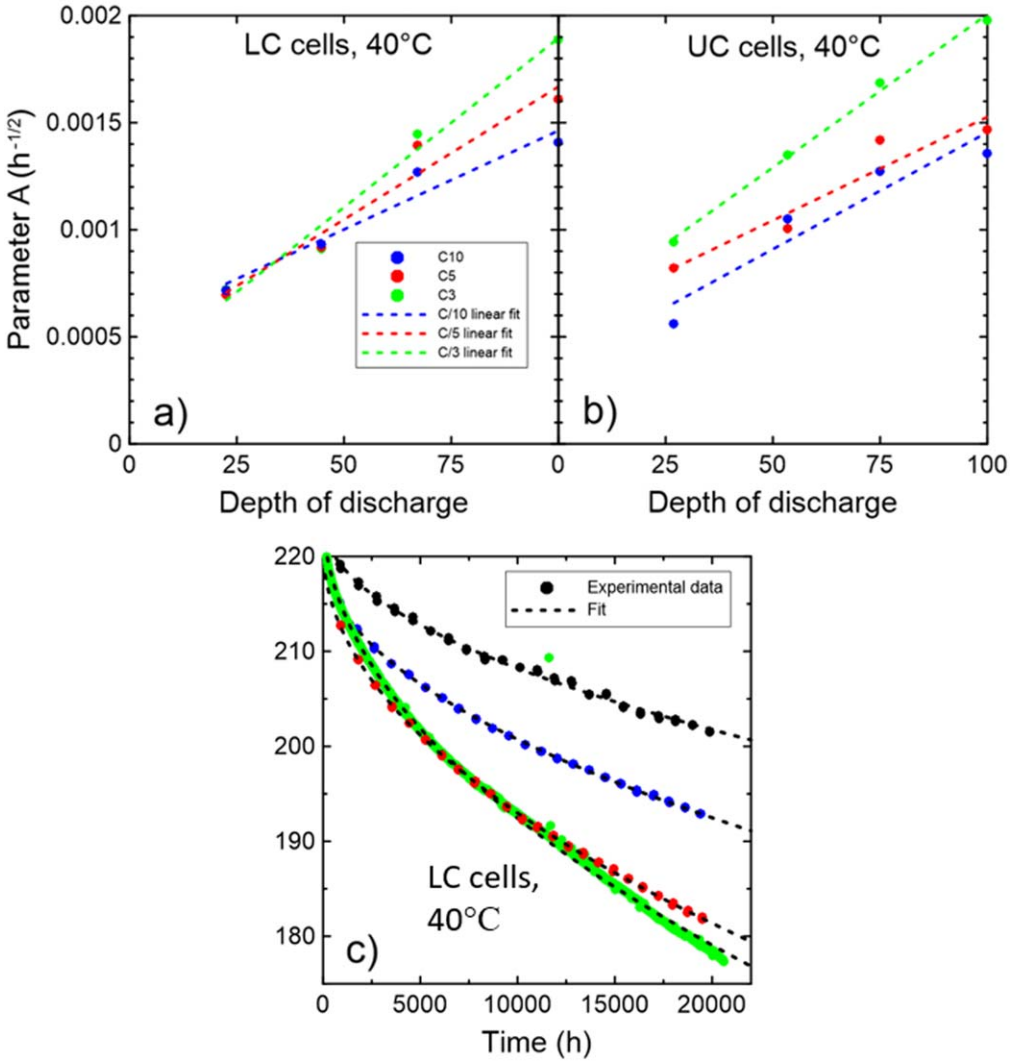


Figure 14. Results of fitting (a) LC cells at 40 °C, (b) UC cells at 40 °C using Eqs. 4 or 6 to obtain parameter A. Equation 4 was used in (a) and (b) at C/10. The Q_{cv} term was neglected in every case. Equation 6 was used in subfigures a) and (b) at C/5 and C/3. Note that the impedance term was negligible at C/10. C/10 data is in blue, C/5 data is in red and C/3 data is in green. Subfigure (c) shows an example of fitting the LC cells with Eq. 4 at 40 °C and C/10.

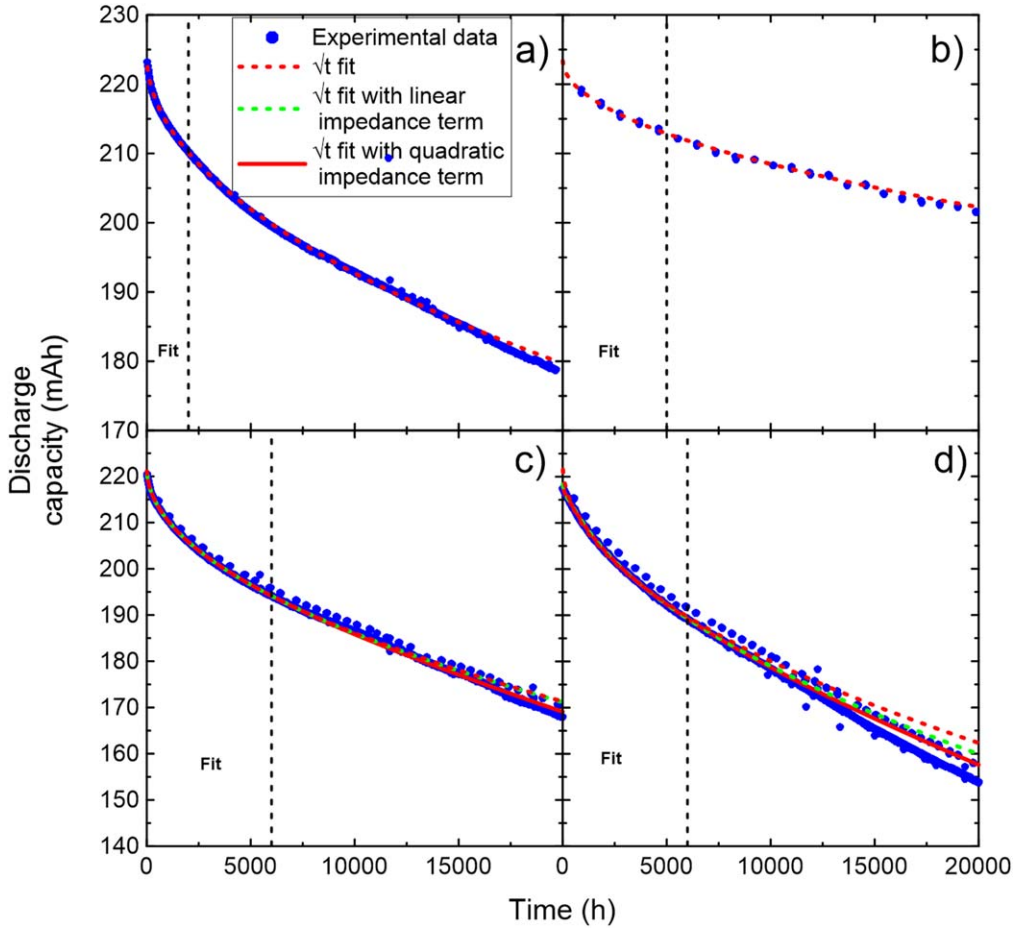


Figure 15. Fitting results for the capacity vs time of LC cells that cycled at 40 °C. Cells with the following cycling conditions have been fitted: a) 0%–100% SOC C/10, b) 0%–25% SOC C/10, c) 0%–100% SOC C/5 and d) 0%–100% SOC C/3. Results for the square root of time model (red dashed lines), for the square root of time models that include the linear impedance term $\Delta V(t) = Ct + D$ that was fitted to the first 6000 h (green dashed lines) and for the square root of time models that include a quadratic impedance term $\Delta V(t) = Et^2 + Ft + G$ that was fitted to the entire 20000 h of data (red solid lines) are included and compared to experimental data (blue line). A black vertical dashed line is used to indicate the maximum time used in the fitting of the model to experiment. The model calculation was then projected to times greater than that of the dashed vertical line.

Table III. Results from the fitting of capacity vs time for LC cells at 40°C in Fig. 15 using Eqs. 4 and 6. (NI) means no impedance term was used in the fit (Eq. 4) and (QI) means that a quadratic impedance term was used in the fit (Eq. 6).

Cell protocol and fitting type	Q_0 (mAh)	A ($h^{-1/2}$)
100% DOD C/10 (NI)	224.2	0.001404
25% DOD C/10 (NI)	223.5	0.0006718
100% DOD C/5 (NI)	221.8	0.001610
100% DOD C/5 (QI)	221.3	0.001659
100% DOD C/3 (NI)	222.8	0.001918
100% DOD C/3 (QI)	216.0	0.002088

Figs. 15c and 15d and was neglected for Figs. 15a and 15b. $\left.\frac{dQ}{dV}\right|_L + \left.\frac{dQ}{dV}\right|_U = 225$ mAh V^{-1} was used for Figs. 15c and 15d. This value can be determined from Fig. 11c.

What is causing cell degradation and how does it depend on DOD, C-rate and temperature?.—Next, our goal was to determine what kind of degradation was occurring in these cells and how the degradation depended on DOD, C-rate and temperature. dV/dQ analysis is a useful method to determine how much shift loss and active mass loss are occurring in a cell.[26,27,53,54] Figure 16 shows the result of a 4D dV/dQ vs Q scan, using the automatic dV/dQ analysis program, of UC and LC cells that cycled for ~20000 h at 40 °C. The positive electrode active mass and the lithium inventory loss Q_{LIL} are plotted as a function of time. Figures 16a and 16c show that UC cells have a bigger positive electrode active mass loss, Δm_p, than LC cells at low DOD. This suggests that positive active mass loss occurs more predominantly at a higher voltage. At high DOD, UC and LC cells have more similar positive active mass losses. Figures 16b and 16d show that the lithium inventory losses are very similar between UC and LC cells for all SOC ranges. This suggests that Q_{LIL} is independent of the average voltage of the SOC ranges, at least for the voltage ranges of the current study. One can find the results for the negative active mass m_n and slippage Δ as a function of time in Fig. S2 as well as some of the dV/dQ fits in Figs. S16 to S21.

Next, manual dV/dQ analysis was done on most of our cells after ~20000 h of cycling using data from the UHPC charger. Figure 17 shows that Δm_p increases with DOD and C-rate for both UC and LC cells at 40 °C but the positive active mass loss is much smaller at 20 °C for all DOD and C-rates. Again, the positive electrode active mass loss is higher for UC cells tested at 40 °C. Similarly, Q_{LIL} increases with DOD and C-rate for cells cycling at 40 °C, while the increases in Q_{LIL} are much smaller at 20 °C as DOD increases. The results shown in Fig. 17, as well as additional results such as Δ and negative electrode active mass, m_n, can be found in Tables SIII to SIX. To better understand the differences, due to errors, between the lithium inventory loss data from Figs. 17b and 17d, the lithium

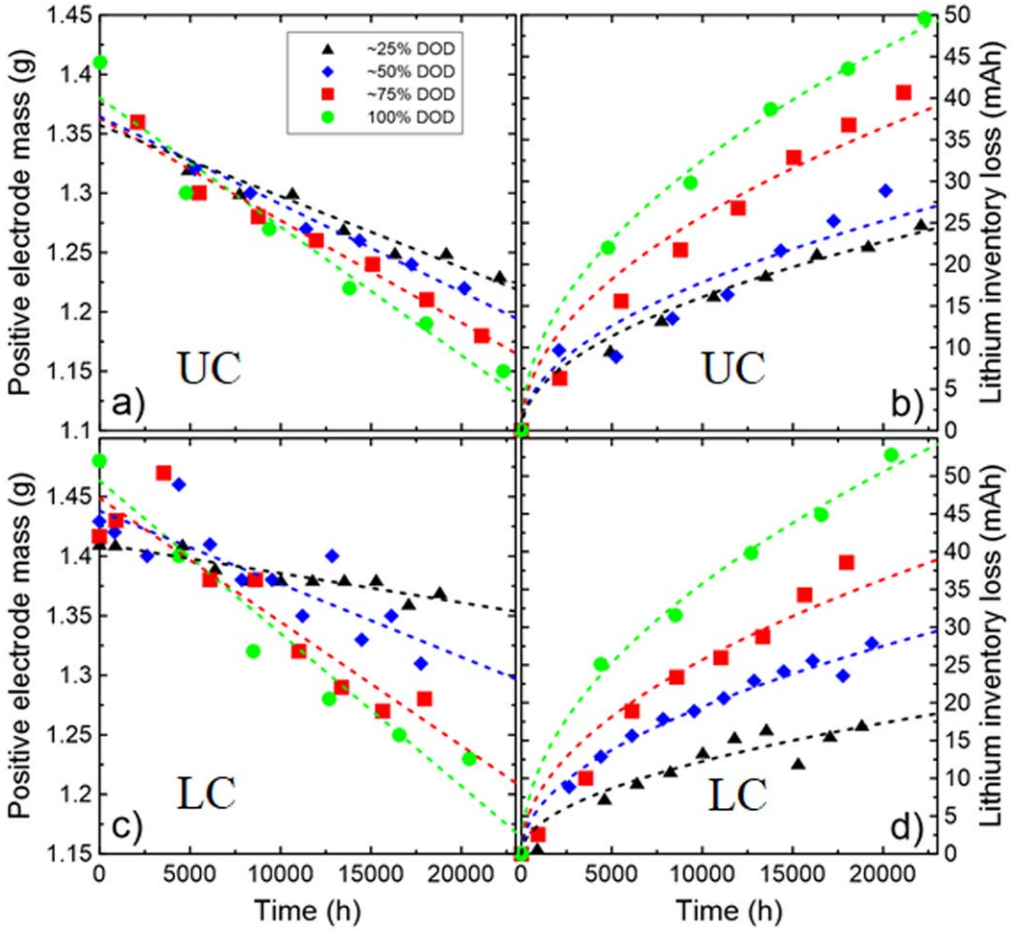


Figure 16. Automatic dV/dQ vs Q scan results for C/10 subfigures (a), (b) UC cells that cycled at 40 °C and subfigures (c), d) LC cells that cycled at 40 °C as a function of cycling time. Subfigures (a) and (c) show positive electrode active mass results and subfigures (b) and (d) show lithium inventory loss results. An automatic 4D dV/dQ vs Q scan was used to get these results. Linear fits (dashed lines in panels (a) and (c)) and square root of time fits (dashed lines in panels (b) and (d)) were applied to the positive electrode active mass and lithium inventory loss data, respectively, to show likely trends.

inventory loss data from Figs. 16b and 16d and the experimental capacity vs time data at C/10 in Figs. S3 to S5, S22 to S23 were made. Figure S22 shows total capacity loss vs lithium inventory loss and Fig. S23 shows lithium inventory loss calculated from Neware charger data (automatic dV/dQ analysis program) vs lithium inventory loss calculated from UHPC charger data (manual dV/dQ analysis). The data in Figs. S22 and S23 are correlated to each other, as expected. The worse error in Fig. S23 is approximately 10 mAh. Note that some errors will be introduced in Fig. S22 due to capacity vs time extrapolation errors. Additional errors can be caused by uncertainties in the initial positive mass m_{p0} and initial slippage Δ_0 of each cell at $t = 0$ h since we only know these values for some of the cells.

For the lithium inventory loss results in Fig. 17, the initial slippage and initial positive mass were fixed to 25 mAh and 1.42 g for all cells as an approximation, using reasonable assumptions. The initial slippage of 25 mAh was found to be the smallest initial slippage in Fig. S2. For the interested reader, state of charge maps of our cells can be found in the supplementary material (see Figs. S24 and S25).[55] Note that these maps show that our cells mostly cycle between fully delithiated graphite and $Li_{0.6}C_6$ at the graphite electrode and between $Li_{0.9}Ni_{0.6}Mn_{0.2}Co_{0.2}O_2$ and $Li_{0.6}Ni_{0.6}Mn_{0.2}Co_{0.2}O_2$ at the positive electrode during checkup cycles (3.0 V–4.1 V).

Next, we wanted to confirm that our cells were anode-limited. Figure 18 shows the positive and negative electrode half-cell V(Q) curves vs Li/Li^+ obtained by manual dV/dQ analysis on UHPC charger data before and after long-term cycling at 40 °C and C/10 for 75%–100% SOC range, 50%–100% SOC range, 25%–100% SOC range and 0%-100% SOC ranges of UC cells. Figure 18 shows that Q_{LIL} increases as a function of DOD and the cells are always anode-limited even as Δm_p increases with DOD.

Most of our cells were showing a volume change of 0.3 ml or less after 20000 h of long-term cycling, so one could wonder if that volume change was caused by gas production or by an increase in jelly roll thickness. This is important to know, since each are the consequence of different degradation modes, and each degradation mode impacts lifetime differently. Figure 19a shows that the cell volume change during cycling, measured by the Archimedes method, is closely related to the thickness change, measured with the linear gage, of the jelly roll. In fact, the experimental data closely follow the predicted volume change (black line) which was calculated by multiplying the thickness change of the jelly roll by the area of the jelly roll (2.0 cm × 3.0 cm). Most cells showed very little, if any, gas in the gas bags which is consistent with all volume change being due to electrode stack thickness growth and not gas production. Cells that cycled at 20 °C seem to have a larger volume change to thickness change slope which may be due to DMC being used instead of EMC in the electrolyte of the cells tested at 20°C, which may result in gas production during cycling.

By pure curiosity, the thickness increase was compared to normalized capacity loss. Figure 19b shows that thickness increase is strongly correlated with the normalized capacity loss during long-term cycling for all types of cells. This is an interesting result that requires further investigation, since it may point toward a possible

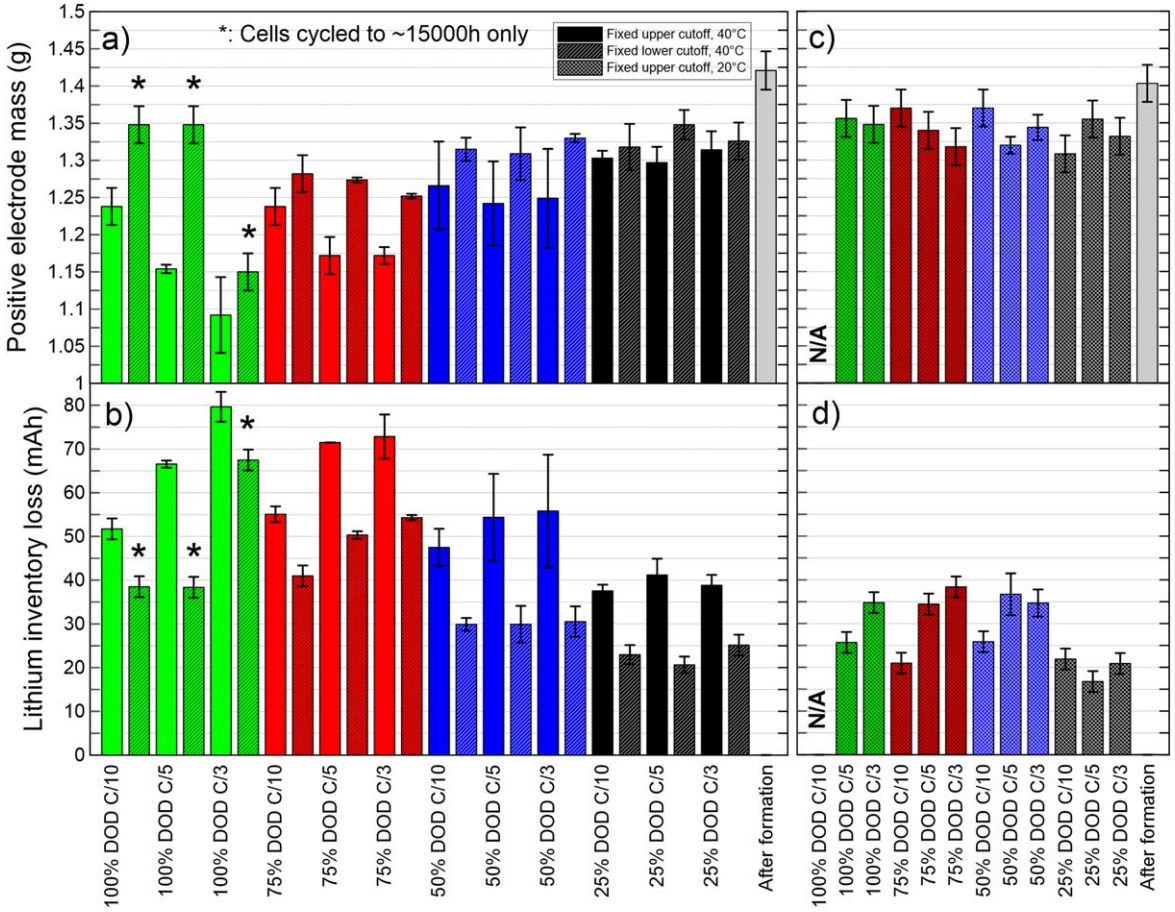


Figure 17. Manual dV/dQ analysis results of UC and LC cells after ~20000 h of cycling, except for the indicated cells (*) which cycled only for ~15000 h. Subfigures (a), (c) Positive electrode mass and subfigures (b), (d) lithium inventory loss is shown. 17 a) and 17 b) show results for cells that cycled at 40 °C, while subfigures (c), (d) show results for cells that cycled at 20 °C. The initial slippage and the initial positive electrode active mass were fixed at 25 mAh and 1.42 g, respectively, for all cells.

direct or indirect effect between the two. However, it is important to remember that correlation doesn't mean causation.

To further investigate, the active mass loss of each electrode was compared to the thickness increase of these cells. Figure 20a shows that the positive electrode active mass loss, from dV/dQ vs Q analysis, and the thickness increase are approximately quadratically related to each other. Negative electrode mass loss also seems to be correlated with thickness increase. However, the active mass loss at the negative electrode is five times less than at the positive electrode. This seems to point toward a positive electrode thickness increase, which will be confirmed later in this work by X-ray computed tomography (CT). We will also confirm, using CT scans, that the thickness of the negative electrode increases during cycling, but much less than the positive electrode. The raw volume, capacity loss and thickness data can be found in Tables SVI to SIX.

While Fig. 20b shows that negative active mass loss occurs in our cells, it shouldn't result in any capacity loss, other than possible lithium inventory loss. In fact, since the cells are balanced to 4.5 V and the maximum voltage of these cells is 4.1 V, the negative electrode has a large excess of graphite and as a result, the negative electrode mass loss is too small to affect capacity loss. Then, due to the large excess of graphite, the lithium coming from the positive electrode during charge will simply find electrically connected graphite particles to intercalate within.

To check if the electrolyte in our cells changed after ~20000 h of cycling, we used Li-ion differential thermal analysis (DTA).[38,56] Differential thermal analysis is straightforward to apply to many cells, very sensitive to electrolyte change and non-destructive. Figure 21 shows DTA results for 100% DOD and ~25% DOD UC cells that cycled at 40 °C compared to the data of a control cell (i.e., no long-term cycling, only formation). Figures S26 to S30 also show DTA results for the other cells. The inverse peak at the highest temperature, approximately −35 °C, occurs at the liquidus point of the electrolyte composition-temperature phase diagram.[38,56–58] The temperature of the liquidus feature is very sensitive to salt concentration and solvent ratios of the electrolyte.[56] Since the temperature of the liquidus feature varies by only 3 °C in Fig. 21 during long-term cycling for both the 100% DOD and the ~25% DOD cell, it can be concluded that changes to the electrolyte are minimal as a function of DOD. In fact, if that shift was entirely caused by salt consumption, it would correspond to approximately 0.1 M of salt consumption.[56] In addition, the DTA traces in Figs. 21 and S26 match quite well to that of a fresh cell after formation, also suggesting minimal changes to the electrolyte. As a comparison, past work by Thompson et al. has shown that when NMC532/graphite cells with 1.1 m $LiPF_6$ EC:EMC 3:7 and 2% VC + 1% DTD are stored at 4.3 V and 55 °C, the area under the curve of the solidus peak at approximatively −60 °C decreases dramatically after 12 months of storage compared to a fresh cell. GCMS of the electrolyte of these cells confirmed a change, compared to a fresh cell, in the fraction of transesterification of ~0.03/month. Note that the fraction of transesterification is defined as the fraction of DMC and DEC molecules created compared to the total amount of linear carbonate molecules. In comparison, for the same electrolyte and cell type, GC-MS showed minimal transesterification at 40 °C and 4.3 V during cycling and storage of around ~0.002/month and the DTA curves had minimal changes similar to that shown in Fig. 21 of this paper. Note that our cells cycled for 2.5 years where those in the

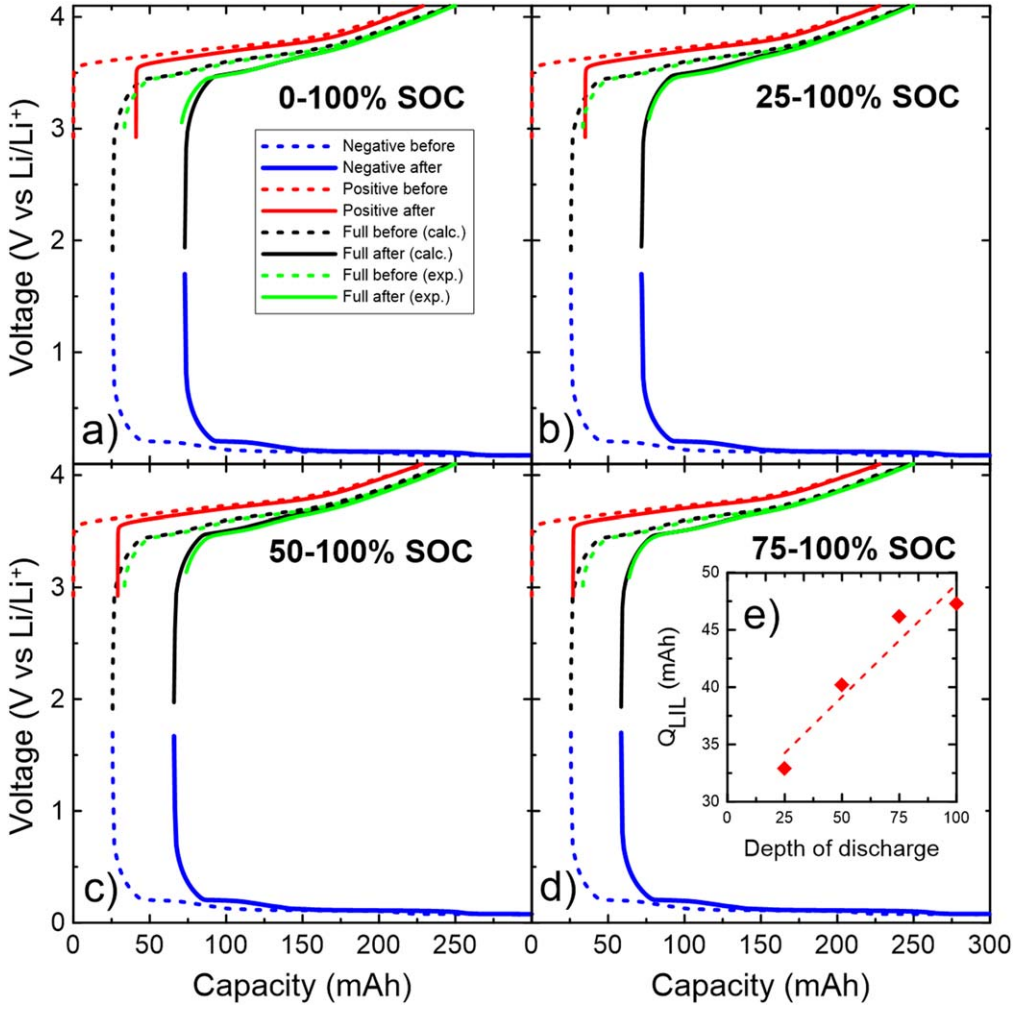


Figure 18. Negative and positive electrode V(Q) curves vs Li/Li^+ before and after long-term cycling (~20000 h) at C/10 and 40 °C of a) 0%–100% SOC range, (b) 25%–100% SOC range, (c) 50%–100% SOC range and d) 75%-100% SOC range for UC cells. Subfigure e) shows lithium inventory loss vs depth of discharge for the cells shown in (a), (b), (c) and (d).

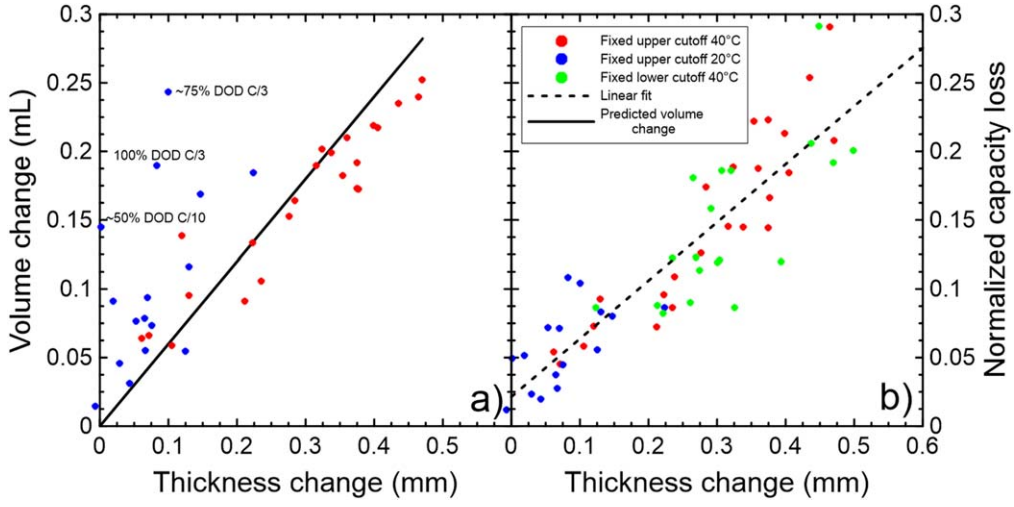


Figure 19. (a) Volume change from Archimedes' principle and (b) normalized capacity loss after ~20000 h of cycling vs thickness change. Red, blue, and green dots indicate UC cells cycling at 40 °C, UC cells cycling at 20 °C and LC cells cycling at 40 °C, respectively. Note that the slope in (b) is 0.42 mm^{-1} and the initial thickness of the cell before long-term cycling was 4.1 mm.

Thompson et al. paper were tested for at most 1 year. As a result, we can confirm that minimal transesterification and minimal salt loss occurred in our cells.

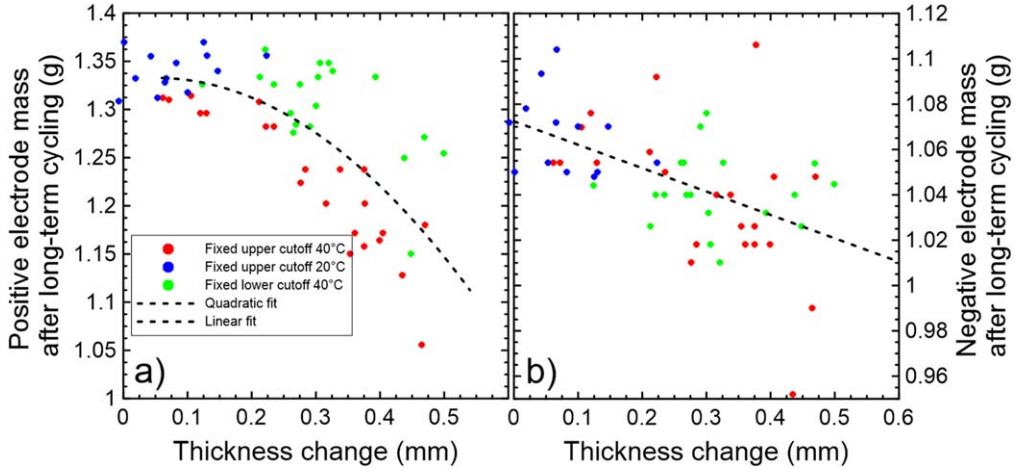


Figure 20. (a) Positive and b) negative electrode active mass after ∼20000 h of cycling vs thickness change of the jelly roll for all UC and LC cells. Red, blue, and green dots indicate UC cells cycling at 40 °C, UC cells cycling at 20 °C and LC cells cycling at 40 °C, respectively. Linear and quadratic fits are also included.

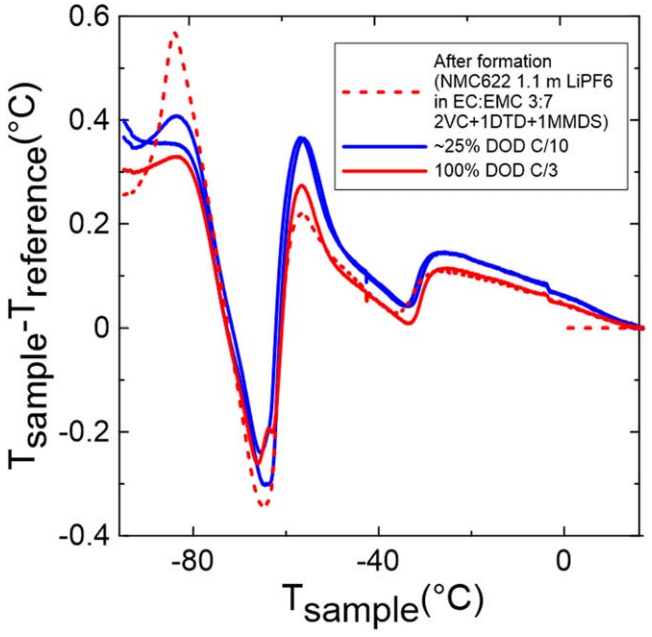


Figure 21. Differential thermal analysis (DTA) vs temperature for UC cells that cycled at 40 °C. The electrolyte originally added to the cells was 1.2 M $LiPF_6$ EC:EMC 3:7 with VC211. The red line shows the results after long-term cycling for the 100% DOD cell and the blue lines show the results after long-term cycling for the ∼25% DOD cell. The red dashed lines show the data for a fresh cell after formation with 1.2 M $LiPF_6$ EC:EMC 3:7 with 2% VC + 1% DTD + 1% MMDS as the electrolyte.

There are, however, some slight changes to the DTA signals as the DOD and C-rate both increase. For example, Figures S28 and S29 show that a double inverse peak at approximately −70 °C is present at high C-rate (C/3 and C/5) and high SOC range (0%–75%, 25%–100% and 0%–100%), which is not seen at low C-rate and lower SOC range. This change can be caused by minimal changes in salt and/or solvent concentration.

Ultrasonic transmission mapping (UTM) is a technique that is straightforward to apply to many cells and is non-destructive. UTM can inform researchers about electrolyte dry-out and empty electrode pores since they result in acoustic impedance mismatch and poor transmission of ultrasonic waves.[31] Electrolyte dry-out can result in impedance growth, so UTM can inform researchers about the mechanisms that result in impedance growth. Figures 22 and S31-S32 summarize some of the data reported above for the cells tested here and provide an ultrasonic 2D image of most of the cells tested. Figure 22 shows the results for UC cells that cycled at 40 °C. Results for the LC cells that cycled at 40 °C and the UC cells that cycled at 20 °C are shown in Figures S31 to S32. Figures 22 and S31–S32 show that cells with large thickness increase (black bars, in mm) also show the poorest transmission to ultrasonic waves (blue = bad ultrasonic transmission, green = medium ultrasonic transmission, red = good ultrasonic transmission). Since volume change and capacity loss correlate with thickness change, they also correlate with the poor transmission. We also see that large C-rate and large SOC range during long-term cycling results in the worse ultrasonic transmission. While it is harder to see any correlation due to missing data in Fig. S32, it is thought that transmission is better at 100% DOD C/3 for cells that cycled at 20 °C than at 40 °C since the cell thickness increase is much lower. The curious reader can have a look at the impact of cell voltage on ultrasonic transmission in Figs. S37 and S38.

The results in Figs. 22 and S31 and S32 can be understood based on the work by Deng et al.[31] That work proposed that cells that have large electrode thickness increase do not have enough electrolyte to fill the increased pore volume. This "unwetting" leads to poor ultrasonic transmission.

To determine at what electrode the increase in thickness is mostly coming from in our cells, some of our cells were sent to the Canadian Light Source synchrotron facility. Figure 27 shows X-ray computed tomography scans of a fresh cell and a UC cell that cycled for 20000 h at ∼75% DOD and C/3 at 40 °C. The thickness of the double-side coated positive electrode was measured at three positions in the cell and the thickness of the separator plus double-side coated negative electrode was also measured at several locations. The average of each showed that the 75% DOD UC cell that cycled at C/3 had a 39% increase in its positive electrode thickness and a 10% increase in its negative electrode thickness when compared to a control cell that was only charged once (i.e. formation). This agrees with Fig. 20, which showed that thickness increase and positive active mass loss were correlated. The increase in electrode thickness with cycling also appears to have filled in the "empty" region at the center of the jelly roll. X-ray CT scans on more cells from this study will be carried out in future work.

The cell cycled at 40 °C with a 25%–100% SOC range described in Fig. 23 also had a large positive electrode active mass loss of 16 percent as shown in Table SIII (assuming an initial positive active mass of 1.4 g). It is likely that the positive electrode active mass loss and positive electrode thickness increase are related, due to

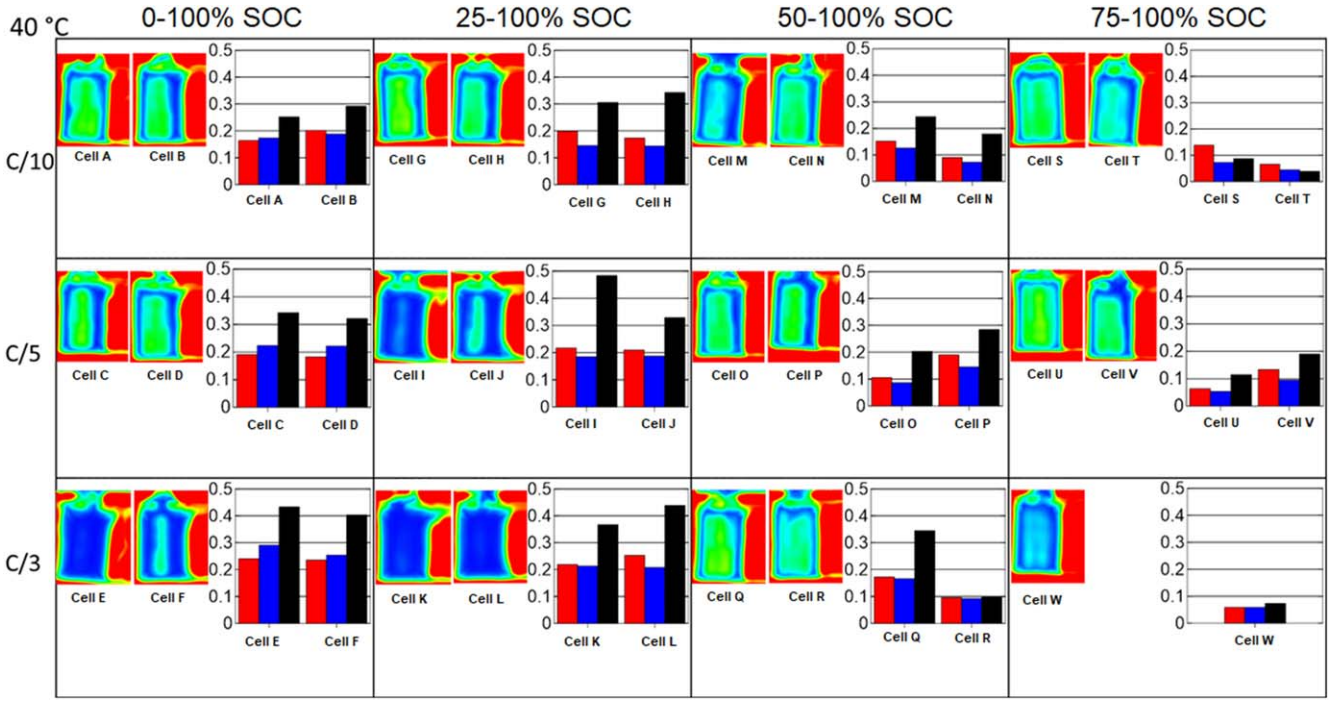


Figure 22. Ultrasonic transmission results for UC cells that cycled at 40 °C for 20000 h. Columns represent cells that cycled over different SOC ranges, while rows represent cells that cycled at different C-rates. In the ultrasonic transmission maps, blue = poor ultrasonic transmission, green = medium ultrasonic transmission, red = good ultrasonic transmission. Poor ultrasonic transmission occurs when the electrolyte does not fully wet the pore spaces in the electrodes. The bar graphs show, in order, the true volume change in ml from Archimedes principle (red), the normalized capacity loss (blue) and the change in thickness in mm (black). When available, the results from duplicate cells are shown. In the ultrasonic color maps, red represents no attenuation, green represents high transmission and blue represents low transmission.

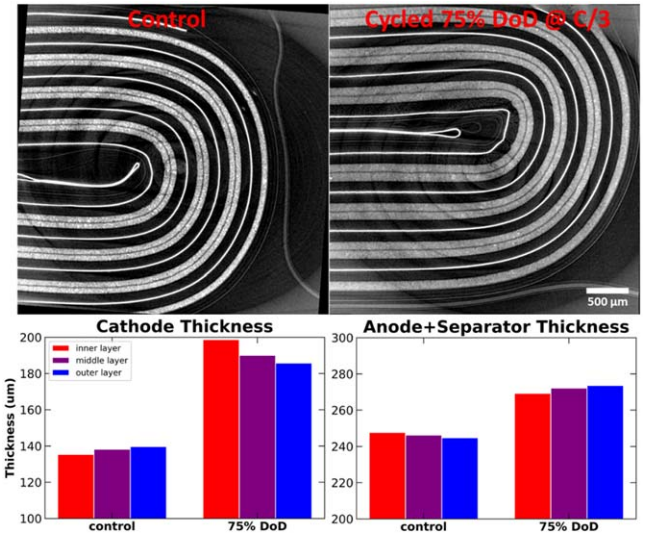


Figure 23. The upper left side of the figure shows a CT-scan of a NMC622/NG that only went through formation (control cell), while the upper right side of the figure shows a CT-scan of a UC cell that cycled at between 25 and 100% SOC at C/3 and 40 °C for 20000 h. The CT scans are presented as absorption images, where the graphite anode and separator are dark (low X-ray absorption) while the copper current collector between the two graphite layers shows as a bright line (high X-ray absorption). The positive electrode layers are relatively bright and the aluminum foil between them is less bright. The lower part of the figure shows the cathode thickness and anode + separator thickness of the same control and ~75% DOD C/3 cells. As mentioned in the main text, most of the thickness increase comes from the cathode (39% increase for the cathode vs 10% for the anode).

disconnection of active particles from the current collector, "electrode breathing" during charge and discharge and microcracking.

Since UTM has suggested that some regions of the jelly roll of some of our cells were not properly wetted after 20000 h of testing, we wanted to see if we could detect the electrolyte in our cells using X-rays. Figure 24 shows X-ray CT scans focused at the edge of the jelly roll to examine the amount of electrolyte outside the jelly roll. Figure 24 shows that, after 20000 h of long-term cycling at 40 °C over a large SOC range, no electrolyte is visible outside the jelly roll. By contrast, some electrolyte can be observed outside the jelly roll for cells that cycled over a 25% SOC range, where cell thickness expansion was less. This explains why the transmission of ultrasonic waves is poor for cells that cycled the most times (C/3 and C/5) over large SOC ranges at 40°C, since there may not be enough electrolyte available in some regions of the jelly roll and/or within the electrode pore spaces.[31]

The lower amount of visible electrolyte in cells that cycled at high SOC range and high C-rate in Fig. 24 can be explained as follows. The increase in electrode thicknesses causes the pore volume inside the electrodes to increase. The electrolyte then fills as much of the empty space inside the electrodes as possible like water in a sponge, reducing the amount of visible electrolyte in X-ray CT images.

Discussion

In this work, lithium inventory loss was the major cause of capacity loss at low C-rate in our anode-limited cells and impedance-related capacity loss was never larger than 2 mAh (1%) at C/10 after 20000 h of cycling. It was shown that the DB model, Eq. 2, was not accurate at describing the capacity loss dependence on DOD and C-rate for the cells tested here. The model was at its worst when trying to predict capacity loss at high C-rate and low DOD for the case of LC cells. Figures 8, 14, 15a and 15b showed that a simple empirical model based on the square root of time model, Eqs. 4 and 5, was more accurate at describing the data at C/10. This model was able to describe the data by assuming that the parameter A was linear vs

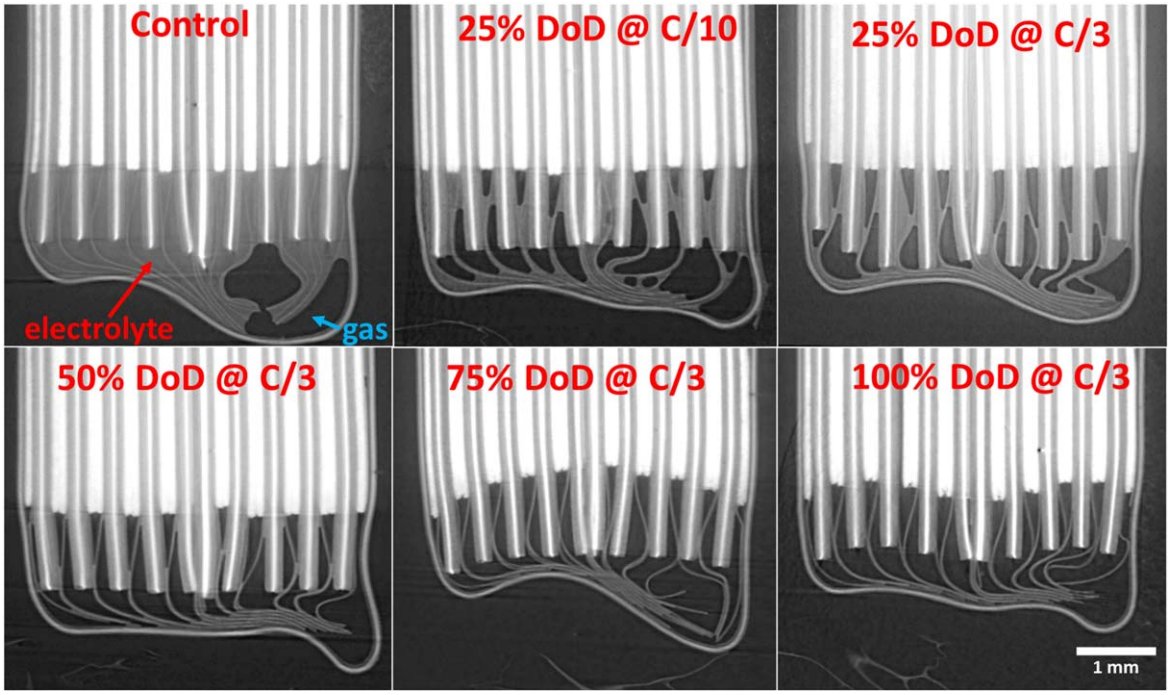


Figure 24. CT-scans of a control cell (immediately after formation), UC cells that cycled at 100%-75% SOC C/10, at 100%-75% SOC C/3, at 100%-50% SOC C/3, at 100%-25% SOC C/3 and 100%-0% SOC C/3. The electrolyte appears as light gray regions outside the jelly roll and gas or vacuum appears as black regions. The white regions represent the electrode of the cells. The plastic casing of the cells and the separators extending beyond the electrodes are also seen as light grey.

DOD. At C/5 and C/3, it was shown that an impedance term needed to be included to account for the capacity loss that resulted from impedance growth. While the impedance term used in this work doesn't account for non-ohmic contributions, it does give a useful first-order approximation of capacity loss due to impedance growth. Interestingly, after fitting the ΔV vs time data of the checkup cycles with a linear model of the form $C\ t + D$, it was shown that C was linear vs C-rate, at least for UC cells that cycled at 40 °C, and approximatively quadratic vs DOD and D was a constant. This could allow researchers to develop an empirical model that describes capacity loss vs time for different DOD and C-rates. However, it is expected that the model will need to be updated in some situations (e.g. high C-rate). This work also partially confirms our hypothesis that ΔV should be Ohmic,at least for the UC cells that cycled at 40 °C. Furthermore, the impedance model developed in this work assumes that impedance growth affects cell voltage curves like a resistor in series ($V(Q) = V_0(Q) \pm \Delta V/2$), where V(Q) is the final voltage curve and $V_0(Q)$ is the initial voltage curve. Of course, the voltage curve also changes because of slippage and active mass loss. This could be a good approximation when ΔV growth is small. However, when ΔV growth is large, it is expected that impedance-related capacity loss would become non-ohmic/non-linear ($V(Q) \neq V_0(Q) \pm \Delta V/2$). The linearity of parameter C in Fig. 13a seems to show that $V(Q) = V_0(Q) \pm \Delta V/2$ is a good approximation for UC cells that cycled at 40 °C in this work. It is possible that the reason why the prediction in Fig. 15d was off by 5 mAh was that the non-ohmic contributions to the cell impedance were not negligible or lithium inventory loss no longer followed the square root of time model for the LC cells that cycled at high C-rate.

As predicted, the capacity fade was worse at higher C-rate and higher DOD. Additionally, the cells were shown to be anode-limited. Interestingly, even though graphite volume vs SOC is flat at high SOC due to the stage 2 to 2 L transition of graphite and rapidly increasing at low SOC, as shown in Fig. 3a, the capacity loss was virtually the same for UC and LC cells that cycled at ~25% and ~50% DOD, as shown in Figs. 7, 16b and 16d. This seems to contradict the DB model that proposed that SEI cracking, caused by graphite volume change, was the cause of SEI growth.

Active mass loss at the positive electrode was shown to be larger at high DOD, high C-rate and high average SOC, likely due to larger lattice volume change at high DOD, more cycles at high C-rate and larger NMC lattice instability at high voltage. It is possible that active mass loss is caused by the polycrystalline nature of the NMC622 that was used. Interestingly, more active mass loss and larger thickness increase were occurring at the positive electrode compared to the negative electrode as shown by Figs. 16, 17, S2, 20 and 23. This is surprising since Fig. 3a suggests the opposite should happen. Additionally, this is equally surprising since natural graphite also has a large irreversible volume increase during cycling.[25]

To test whether the positive electrode thickness increase was due to the polycrystalline nature of the NMC622 used, a single crystal NMC532/artificial graphite (AG) cell with 1.5 M $LiPF_6$ EC:EMC 3:7 and 2% VC + 1% DTD was cycled for 20000 h at C/3 between 3.0 V and 4.1 V. The cell in question is a STO0 cell from J. H. Cheng et al.[53] The cell in question was then sent to the Canadian light source for an X-ray CT scanning. Similarly, a control cell that was under storage for 20000 h was also scanned. The results of the CT scans are shown in Fig. S35. From the data, we can conclude that during long-term cycling the anode thickness expanded by 7% and the cathode thickness expanded by 11%. This means that the positive electrode of the single crystal NMC532/AG cell expanded 3.5 times less than the positive electrode of the polycrystalline NMC622/NG cells, even though the polycrystalline cell was cycling at lower DOD than the single-crystal cell, while the negative electrode thickness remained virtually the same. Additionally, from the work of J. H. Cheng et al., the single crystal NMC532/AG cell had virtually no active mass loss at the positive electrode after ~20000 h of cycling. This suggests, again, that active mass loss at the positive electrode and thickness growth are related to each other. Additionally, the increase in cell thickness of the polycrystalline NMC622/NG cells was worse at high DOD, as shown by Tables SVI to SVIII.

It is thought that the impedance growth in the cells described in this work is mostly caused by positive electrode active mass loss,

thickness growth at the positive electrode and "unwetting" of the jelly roll, as Figs. 22 and S36 might suggest.

Finally, the electrolyte of those cells after ~20000 h of cycling did not change dramatically. If salt consumption occurred in those cells, the salt consumption should be on the order of 0.1 M of $LiPF_6$ and the change in the fraction of transesterification during cycling is thought to be in the order of ~0.002/month or less.

Conclusions

The above work can be summarized by the following conclusions:

- Capacity loss in these polycrystalline NMC622/NG pouch cells results primarily from lithium inventory loss. Even though there is positive electrode active mass loss in those cells, they remain anode-limited. As a result, the positive active mass loss does not directly contribute to capacity fade.
- Capacity vs time predictions can be improved by adding a first-order impedance term to a simple $t^{1/2}$ lithium inventory loss model.
- Lithium inventory loss increases when the SOC range increases.
- Lithium inventory loss is worse at high temperature. However, UC cells and LC cells with the same SOC range have similar lithium inventory loss.
- Impedance growth and impedance-related capacity loss are worse at higher temperature (40 °C vs 20 °C) and higher DOD.
- Positive active mass loss is worse at high temperature, at higher DOD and at higher average SOC.
- The correlation between capacity loss and electrode thickness increase is interesting and requires further work.
- The correlation between positive active mass loss, jelly roll thickness increase and impedance growth is interesting and requires further work.
- X-ray CT-scans show that most of the jelly roll thickness increase comes from the positive electrode and this thickness increase is worse at high DOD.
- Our work contradicts the model of Deshpande and Bernardi in its current form.[8] It is still possible that SEI cracking could occur at negative electrodes, but the DB model would need to be adjusted and more experiments would be needed to confirm or not if SEI cracking occurs.
- The increase in positive electrode thickness causes the pore volume inside the electrodes to increase. The electrolyte then fills as much of the empty space inside the electrodes as possible like a sponge, explaining why no electrolyte is observed outside the jelly roll for the cells that cycled over a large SOC range. If stack expansion continues further, this will eventually cause, presumably dramatic, cell failure by "unwetting."
- Poor ultrasonic transmission at large SOC ranges and C-rates can be explained by the lower amount of electrolyte available to fill the expanded electrode pore spaces and/or lower amount of electrolyte available in some regions of the jelly roll.
- Since the temperature of the liquidus feature does not vary dramatically for cells during cycling and the DTA traces of most cells match the DTA traces of fresh cells after formation, it can be concluded that changes to the electrolyte are minimal as a function of DOD and C-rate under the conditions probed here. More specifically, if any salt consumption is present in those cells, it should be in the order of a 0.1 M of $LiPF_6$ loss after ~20000 h of cycling and the change in the fraction of transesterification during cycling should be in the order of ~0.002/month or less.

In our opinion, the NMC622/NG cells presented here do not represent the state of the art in terms of capacity retention vs cycle number or time. They are less impressive than the single crystal NMC532/AG (artificial graphite) cells described by Harlow et al.[26] and by Cheng et al.[53] There are two main reasons for this. First, the polycrystalline NMC622 positive electrode swells substantially during extended cycling and this is worse as the SOC range is increased. By contrast, as shown in Fig. S35, the single crystal NMC532 electrode shows virtually no swelling after the same number of cycles under the same conditions. Second, the natural graphite electrode material used in the NMC622/NG cells has been shown to be inferior to the artificial graphite used in the single crystal NMC532/AG cells in terms of swelling and rate of Li inventory loss.[25] However, the polycrystalline NMC622/NG cells shown here still perform extremely well at low DOD despite their lower performance than the single-crystal cell at large DOD.

Acknowledgments

The authors acknowledge the financial support of NSERC and Tesla Canada under the auspices of the Industrial Research Chair program. A. Louli acknowledges scholarship support from NSERC.

ORCID

Roby Gauthier https://orcid.org/https://orcid.org/0000-0003-4033-8578
Aidan Luscombe https://orcid.org/https://orcid.org/0000-0003-0552-4930
Toby Bond https://orcid.org/https://orcid.org/0000-0003-2043-0094
Michel Johnson https://orcid.org/0000-0001-6445-2813
Jessie Harlow https://orcid.org/0000-0002-3928-2573
A. J. Louli https://orcid.org/0000-0001-9819-1324
J. R. Dahn https://orcid.org/0000-0002-6997-2436

References

1. J. S. Edge et al., *Phys. Chem. Chem. Phys.*, **23**, 8200 (2021).
2. S. J. An, J. Li, C. Daniel, D. Mohanty, S. Nagpure, and D. L. Wood, *Carbon*, **105**, 52 (2016).
3. H. J. Ploehn, P. Ramadass, and R. E. White, *J. Electrochem. Soc.*, **151**, A456 (2004).
4. A. J. Smith, J. C. Burns, D. Xiong, and J. R. Dahn, *J. Electrochem. Soc.*, **158**, A1136 (2011).
5. A. Wang, S. Kadam, H. Li, S. Shi, and Y. Qi, *NPJ Comput. Mater.*, **4**, 1 (2018).
6. F. Single, A. Latz, and B. Horstmann, *ChemSusChem*, **11**, 1950 (2018).
7. K. Ushirogata, K. Sodeyama, Z. Futera, Y. Tateyama, and Y. Okuno, *J. Electrochem. Soc.*, **162**, A2670 (2015).
8. R. D. Deshpande and D. M. Bernardi, *J. Electrochem. Soc.*, **164**, A461 (2017).
9. L. K. Willenberg, P. Dechent, G. Fuchs, D. U. Sauer, and E. Figgemeier, *Sustainability*, **12**, 557 (2020).
10. Y. Preger, H. M. Barkholtz, A. Fresquez, D. L. Campbell, B. W. Juba, J. Romàn-Kustas, S. R. Ferreira, and B. Chalamala, *J. Electrochem. Soc.*, **167**, 120532 (2020).
11. L. Willenberg, P. Dechent, M. Eckert, D. Sauer, and E. Figgemeier, *Systematic Investigation of Mechanical Ageing of High Energy Lithium-Ion Batteries* (2019).
12. X. M. Liu and C. B. Arnold, *J. Electrochem. Soc.*, **163**, A2501 (2016).
13. A. Maheshwari, M. Heck, and M. Santarelli, *Electrochim. Acta*, **273**, 335 (2018).
14. J. Wang, P. Liu, J. Hicks-Garner, E. Sherman, S. Soukiazian, M. Verbrugge, H. Tataria, J. Musser, and P. Finamore, *J. Power Sources*, **196**, 3942 (2011).
15. E. Sarasketa-Zabala, I. Gandiaga, E. Martinez-Laserna, L. M. Rodriguez-Martinez, and I. Villarreal, *J. Power Sources*, **275**, 573 (2015).
16. S. Watanabe, M. Kinoshita, T. Hosokawa, K. Morigaki, and K. Nakura, *J. Power Sources*, **260**, 50 (2014).
17. J. Li, J. Harlow, N. Stakheiko, N. Zhang, J. Paulsen, and J. Dahn, *J. Electrochem. Soc.*, **165**, A2682 (2018).
18. Y. Zhang, R. Xiong, H. He, X. Qu, and M. Pecht, *Appl. Energy*, **255**, 113818 (2019).
19. J. de Hoog, J.-M. Timmermans, D. Ioan-Stroe, M. Swierczynski, J. Jaguemont, S. Goutam, N. Omar, J. Van Mierlo, and P. Van Den Bossche, *Appl. Energy*, **200**, 47 (2017).
20. Y. Gao, J. Jiang, C. Zhang, W. Zhang, and Y. Jiang, *J. Power Sources*, **400**, 641 (2018).
21. M. Lewerenz, J. Münnix, J. Schmalstieg, S. Käbitz, M. Knips, and D. U. Sauer, *J. Power Sources*, **345**, 254 (2017).
22. Y. Cui, C. Du, G. Yin, Y. Gao, L. Zhang, T. Guan, L. Yang, and F. Wang, *J. Power Sources*, **279**, 123 (2015).
23. K. Darcovich, S. Recoskie, and F. Fattal, *J. Appl. Electrochem.*, **50**, 111 (2020).
24. D. Y. Wang et al., *J. Electrochem. Soc.*, **161**, A1818 (2014).
25. S. L. Glazier, J. Li, A. J. Louli, J. P. Allen, and J. R. Dahn, *J. Electrochem. Soc.*, **164**, A3545 (2017).
26. J. E. Harlow et al., *J. Electrochem. Soc.*, **166**, A3031 (2019).
27. H. M. Dahn, A. J. Smith, J. C. Burns, D. A. Stevens, and J. R. Dahn, *J. Electrochem. Soc.*, **159**, A1405 (2012).
28. A. J. Louli, J. Li, S. Trussler, C. R. Fell, and J. R. Dahn, *J. Electrochem. Soc.*, **164**, A2689 (2017).

29. L. Willenberg, P. Dechent, G. Fuchs, M. Teuber, M. Eckert, M. Graff, N. Kürten, D. U. Sauer, and E. Figgemeier, *J. Electrochem. Soc.*, **167**, 120502 (2020).
30. T. Bond, J. Zhou, and J. Cutler, *J. Electrochem. Soc.*, **164**, A6158 (2016).
31. Z. Deng, Z. Huang, Y. Shen, Y. Huang, H. Ding, A. Luscombe, M. Johnson, J. E. Harlow, R. Gauthier, and J. R. Dahn, *Joule*, **4**, 2017 (2020).
32. L. M. Thompson, J. E. Harlow, A. Eldesoky, M. K. G. Bauer, J. H. Cheng, W. S. Stone, T. Taskovic, C. R. M. McFarlane, and J. R. Dahn, *J. Electrochem. Soc.*, **168**, 020532 (2021).
33. A. J. Louli, L. D. Ellis, and J. R. Dahn, *Joule*, **3**, 745 (2019).
34. L. Ma et al., *ACS Appl. Energy Mater.*, **1**, 7052 (2018).
35. S.-T. Myung, K. Izumi, S. Komaba, Y.-K. Sun, H. Yashiro, and N. Kumagai, *Chem. Mater.*, **17**, 3695 (2005).
36. T. M. Bond, J. C. Burns, D. A. Stevens, H. M. Dahn, and J. R. Dahn, *J. Electrochem. Soc.*, **160**, A521 (2013).
37. MATLAB, *9.8.0.1417392 (R2020a)* (The MathWorks Inc., Natick, Massachusetts) (2020).
38. R. P. Day, J. Xia, R. Petibon, J. Rucska, H. Wang, A. T. B. Wright, and J. R. Dahn, *J. Electrochem. Soc.*, **162**, A2577 (2015).
39. T. W. Wysokinski, D. Chapman, G. Adams, M. Renier, P. Suortti, and W. Thomlinson, *Nucl. Instrum. Methods Phys. Res., Sect. A*, **775**, 1 (2015).
40. M. Vogelgesang, T. Farago, T. F. Morgeneyer, L. Helfen, T. dos Santos Rolo, A. Myagotin, and T. Baumbach, *J Synchrotron Rad*, **23**, 1254 (2016).
41. R. Deshpande, M. Verbrugge, Y.-T. Cheng, J. Wang, and P. Liu, *J. Electrochem. Soc.*, **159**, A1730 (2012).
42. W. Lee, S. Muhammad, T. Kim, H. Kim, E. Lee, M. Jeong, S. Son, J.-H. Ryou, and W.-S. Yoon, *Adv. Energy Mater.*, **8**, 1701788 (2018).
43. J. R. Dahn, *Phys. Rev. B*, **44**, 9170 (1991).
44. P. Yan, J. Zheng, M. Gu, J. Xiao, J.-G. Zhang, and C.-M. Wang, *Nat. Commun.*, **8**, 14101 (2017).
45. B. Gyenes, D. A. Stevens, V. L. Chevrier, and J. R. Dahn, *J. Electrochem. Soc.*, **162**, A278 (2014).
46. A. J. Smith, J. C. Burns, X. Zhao, D. Xiong, and J. R. Dahn, *J. Electrochem. Soc.*, **158**, A447 (2011).
47. L. D. Ellis, J. P. Allen, I. G. Hill, and J. R. Dahn, *J. Electrochem. Soc.*, **165**, A1529 (2018).
48. P. M. Attia, W. C. Chueh, and S. J. Harris, *J. Electrochem. Soc.*, **167**, 090535 (2020).
49. X. Ma, J. E. Harlow, J. Li, L. Ma, D. S. Hall, S. Buteau, M. Genovese, M. Cormier, and J. R. Dahn, *J. Electrochem. Soc.*, **166**, A711 (2019).
50. M. Doyle, T. F. Fuller, and J. Newman, *J. Electrochem. Soc.*, **140**, 1526 (1993).
51. S. Atlung, K. West, and T. Jacobsen, *J. Electrochem. Soc.*, **126**, 1311 (1979).
52. X.-G. Yang, Y. Leng, G. Zhang, S. Ge, and C.-Y. Wang, *J. Power Sources*, **360**, 28 (2017).
53. J. H. Cheng, J. E. Harlow, M. B. Johnson, R. Gauthier, and J. R. Dahn, *J. Electrochem. Soc.*, **167**, 130529 (2020).
54. W. Dose, C. Xu, C. Grey, and M. Volder, *Cell Reports Physical Science*, **1**, 100253 (2020).
55. J. Christensen and J. Newman, *J. Electrochem. Soc.*, **152**, A818 (2005).
56. M. Bauer, *Thesis* (2018), (https://DalSpace.library.dal.ca//handle/10222/74208).
57. M. S. Ding, K. Xu, and T. R. Jow, *J. Electrochem. Soc.*, **147**, 1688 (2000).
58. M. S. Ding, K. Xu, S. Zhang, and T. R. Jow, *J. Electrochem. Soc.*, **148**, A299 (2001).